



Federal Office
for Information Security

5G Risk Analysis

Detailed Analysis: **Multi-Access Edge Computing (MEC)**



Version History

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1 List of Abbreviations

3GPP	3rd Generation Partnership Project
5GC	5G Core
AF	Application Function
AGV	Automated Guided Vehicles
AMF	Access and Mobility Function
AP	Attack Path
APP	Application
CFS	Customer Facing Service
CPU	Central Processing Unit
CUPS	Control User Plane Separation
CVE	Common Vulnerabilities and Exposures
DC	Data Center
DN	Data Network
DNS	Domain Name Server
DNSSEC	Domain Name System Security Extensions
DoS	Denial of Service
DSGVO	Datenschutz-Grundverordnung
DU	Distributed Unit
EAP-AKA	Extensible Authentication Protocol for UMTS Authentication and Key Agreement
EAP-AKA'	Extensible Authentication Protocol for Authentication and Key Agreement (3GPP successor of EAP-AKA for all 3GPP networks)
ETSI	Europäisches Institut für Telekommunikationsnormen
IMS	IP Multimedia Subsystem
IMSI	International Mobile Subscriber Identity
GTP	GPRS (General Packet Radio Service) Tunnelling Protocol
IPsec	Internet Protocol Security
JSON	JavaScript Object Notation
JWT	JSON Web Token
LAN	Local Area Network
LCM	Life Cycle Management
LI	Lawful Interception
MEC	Multi-Access Edge Computing
MEO	Multi-Access Edge Orchestrator
MFA	Multi Factor Authentication
MNO	Mobile Network Operator

NEF	Network Exposure Function
NRF	Network Repository Function
OpenSSL	Open Secure Sockets Layer
OSS	Operating Support System
OAuth2	Open Authorization
PDU	Protocol Data Unit
PLC	Programmable Logic Controller
PLMN	Public Land Mobile Network
PLMNO	Public Land Mobile Network Operator
PCF	Policy Control Function
RAN	Radio Access Network
RD	Retained Data
SIM	Subscriber Identification Module
SMF	Session Management Function
SUPI	Subscriber Permanent Identifier
Svc	Service Instance
TKG	Telekommunikationsgesetz (Telecommunications Act)
TLS	Transport Layer Security
TPM	Trusted-Platform-Modul
UDM	User Data Management
UE	User Equipment
UPF	User Plane Function
UPS	Uninterruptible Power Supply
USIM	Universal Subscriber Identification/Identity Module
WAN	Wide Area Network
XML	Extensible Markup Language
XXE	XML External Entity attack

2 Introduction and Purpose

2.1 Intention

The intention of the document is to dive deep into the analysis of risk scenarios where the MEC layer is the attacked or failing subsystem of a 5G network.

This document shows the current view on the probability that any risk scenarios could become reality.

The study is only available in English language, as the target group is primarily the international IT security community. The study is also in the context of English publications on this topic.

2.2 Link to Framework Document

This document is linked to the framework document and provides relevant information about **Multi-Access Edge Computing (MEC)** in 5G mobile networks and a probability assessment of relevant risk scenarios described in the framework document. The overall risk assessment for each risk scenario consists of the results of this analysis and the impact described in the framework document. The overall risk assessment for the management system is summarized in Chapter 6 of this document and in the framework document.

3 Methodology

The methodology used for the 5G Risk Analysis (5G-RA) is based on PASTA. In this document and the related documents, we will refer to the methodology as 5G-RA.

This chapter explains the seven steps of the 5G-RA methodology and assigns them to the corresponding PASTA steps.

Step 1: Definition of Risk Scenarios

In the first step of the 5G-TARA, the 5G network needs to be analysed from the following three perspectives:

1. Public Mobile Network Operator (MNO) Perspective,
2. Private 5G Network Operator Perspective,
3. User Perspective.

Several risks with a significant impact on the society or public safety have been identified by the legislative in the past. To mitigate these risks, laws were set into force, e.g. the legislative to ensure emergency calls are available (TKG §165). Most of these laws define fines for mobile network operators if the regulative obligation is not followed.

These perspectives are explained in detail in Chapter 6 of the framework document. The goal is to identify the possible risks that the 5G network might be exposed to from each perspective.

Step 2: 5G System Diagram

In step 2, a detailed system diagram of the parts of a 5G network needs to be created. This diagram will serve as a basis for the following steps.

Step 3: Refinement of Detailed 5G Subsystem Diagrams

In this step, more technical definitions and explanations of the relevant 5G subsystems are introduced. For example, various network functions are part of the 5G core network. This level of detail serves as a basis for the following steps.

Step 4: Collection of Known 5G Threats

In this step, publicly known 5G specific threats are collected. For this purpose, relevant studies and input documents are collected. This is the basis for the identification of threats in the next step.

Step 5: Search of 5G Vulnerabilities

Step 5 includes the analysis and identification of relevant threats for each risk scenario. The findings of this step are part of the corresponding Detailed Risk Analyses.

Step 6: Examination of Identified Threats

In this step, all threats that were found in step 5 are examined on whether there is a possibility for this threat to be realized. Hypothetical scenarios are used as well as references to successful attacks, which were performed in the past.

Step 7: Risk Calculation

The last step of the methodology is a calculation of the risk for all threats that were identified in step 5 and are still considered relevant after step 6 was carried out. The overall results are presented in Chapter 6 of this document.

The following table assigns the steps of the 5G-RA to the steps of PASTA and gives an overview of the document containing the results of each step.

Table 1: Assignment of 5G-RA steps to PASTA-steps

#	5G-RA methodology	Document Layer	PASTA
1	Definition of Risk Scenarios	Framework	Define Objectives
2	5G System Diagram	Framework	Define Technical Scope
3	Refinement of Detailed 5G Subsystem Diagrams	Framework & Detailed Risk Analysis	Application Decomposition
4	Collection of Known 5G Threats	Detailed Risk Analysis	Threat Analysis
5	Search of 5G Vulnerabilities	Detailed Risk Analysis	Vulnerability and Weakness Analysis
6	Examination of the Identified Threats	Detailed Risk Analysis	Attack Modelling
7	Risk Calculation	Framework & Detailed Risk Analysis	Risk and Impact Analysis

4 5G Network Model: Multi Access Edge Computing (MEC)

The 5G network model introduced in the following sections is used to split the 5G-RA into multiple Detailed Risk Analyses. The following model follows standards (like 3GPP standards) as much as possible. The 5G network model comprises a variety of interconnected subsystems that are categorized on a high-level view as shown in Figure 1. In the context of this risk analysis, a 5G Standalone system is assumed. Therefore, the Radio Access Network (RAN) as well as the Core are 5G-specific subsystems.

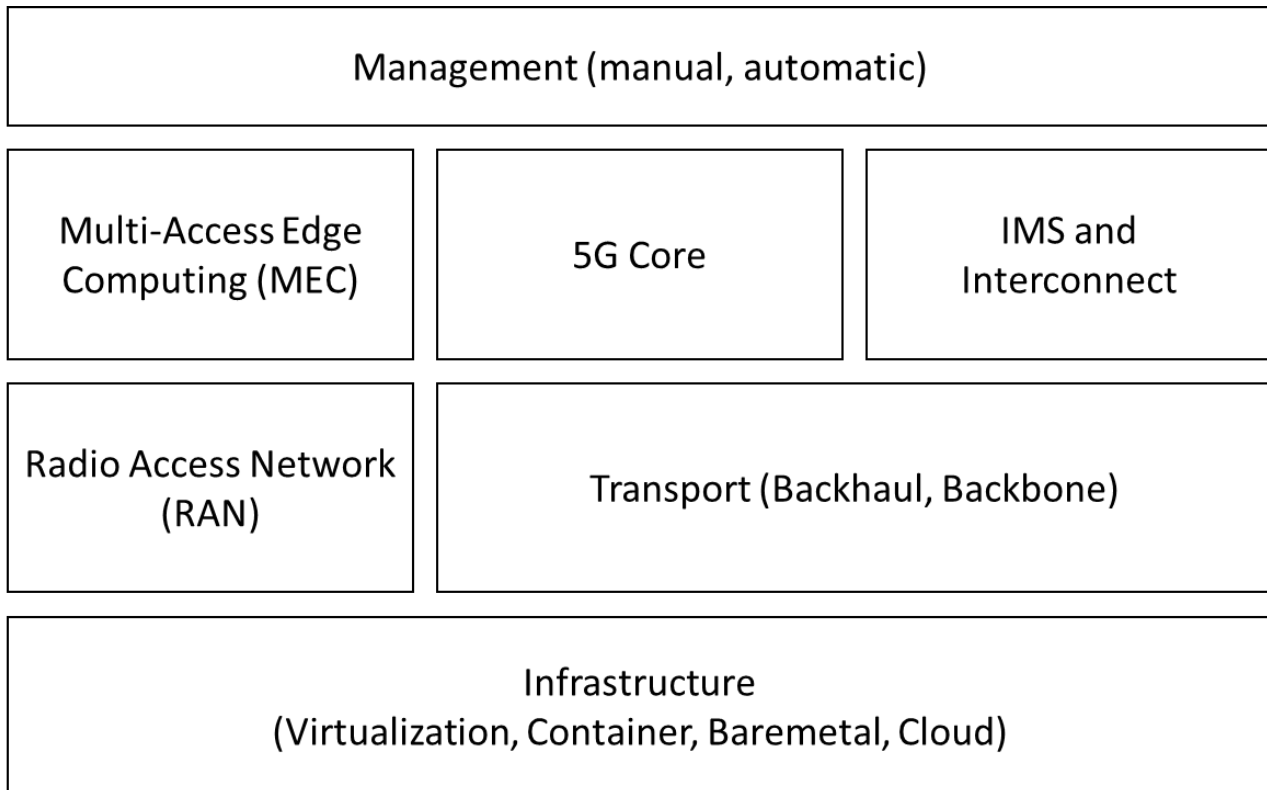


Figure 1: 5G Network Model: High level view

Other subsystems are not specifically designed for the 5G architecture and used either to operate the 5G system or to connect it to other internal and external networks.

In the evolving landscape of 5G mobile communication systems, Multi-Access Edge Computing (MEC) introduces a paradigm where cloud-proven infrastructure is seamlessly integrated with communication networks. This integration not only promises reduced latency and enhanced bandwidth, or information exchange with the cellular network core, but also introduces unique challenges in terms of system architecture and security.

In this chapter, we aim to define the technical scope of the MEC in 5G systems risk analysis, elucidating the system architecture through detailed descriptions of system entities, their interactions, and equivalent terminologies in existing standards from 3GPP, ETSI, and the cloud software stack. As the standards use different terminology, some of which overlap, we first define the terminology used in this report, which is based on a technical, infrastructure and implementation-oriented style. To provide a concrete example for implementation, Kubernetes is selected as the reference system for explanation. However, the terminology is mappable to other technologies as well.

4.1 System Architecture and Entities

4.1.1 Core Entities and Their Implementation

The foundational element of the MEC architecture is the **Service Instance** (short: **Svc**), which represents a standalone software instance, i.e., an instance allocates CPU, storage, and networking resources as required, that performs a specific task. These instances can be scaled horizontally based on load, implying the deployment of multiple instances for load distribution. In a Kubernetes-based environment, a Service Instance is implemented as a Pod or Container. It corresponds to the *MEC APP* in ETSI MEC standards and the *Network Function* (e.g., a Network Exposure Function, NEF) in 3GPP 5G standards.

Service Interface(s) are critical as they define the communication interfaces for Service Instances. These interfaces, often realized as APIs like REST-APIs, do not allocate resources themselves but facilitate the interaction between services. In Kubernetes, this is managed through a Kubernetes Service construct. The ETSI MEC and 3GPP 5G equivalents are *MEC Service* and, as an example, the *Nnef_AnalyticsExposure* API of NEF, respectively.

An **Application** in MEC architecture is conceptualized as a set or chain of services that collectively deliver comprehensive functionality, adhering to the microservices architecture model. This composite structure does not allocate resources independently; rather, it is the individual Service Instance within the application that manage resource allocation. Kubernetes supports such architectures through Deployments, ReplicaSets or similar.

The **Host** and **Cluster** form the physical and logical grouping of resources. A Host, implemented as a Kubernetes Node, is a single server system that houses one or more Service Instances. A Cluster, on the other hand, represents a group of such hosts configured to work as a unified system, managed in Kubernetes as a Kubernetes Cluster.

The **Orchestrator** plays a pivotal role as the management layer that controls the deployment and destruction of Service Instances across hosts or clusters. This entity is related to the *MEC orchestrator* in ETSI MEC standards.

Lastly, the **Platform Application** provides essential system services and interfaces for effective cluster management and service operation, such as DNS services. In Kubernetes, these functionalities are typically managed by services and pods within the “kube-system” namespace, paralleling the *MEC Platform* in ETSI MEC standards.

4.1.2 Terminology Mapping to ETSI MEC

The mapping of ETSI terminology to the report terminology is listed in Table 2. Additionally, the reference MEC ETSI architecture is depicted in Figure 2 providing an overview of the mapping between ETSI and the report terminology.

Table 2: Mapping of ETSI terminology to report terminology

ETSI Terminology	Report Terminology
Device/User/MEC APP	Service Instance
MEC Service	Service Interface
MEC Platform	Platform Application
MEC Host	Host or Cluster
MEC Orchestrator	Orchestrator

4.1.3 Architectural Overviews: Network and Service Platforms

Two system perspectives are used in this analysis and described in this section: the network architecture and the service-platform architecture. The network architecture perspective (Section 4.1.3.1) aims to support the risk analysis from the communication perspective, to analyze reachable hosts and the data path of end user traffic. Complementary, the service-platform architecture perspective (Section 4.1.3.2) supports the analysis of shared platform resources regarding the operation of services. To create a complete picture, the architectures include the main components of a 5G Mobile Network which are slightly faded out in the graphics to focus on the relevant MEC components.

4.1.3.1 Network Architecture

As depicted in Figure 3, the network architecture comprises a central site, referred to as a "Cloud Data Center" (Cloud DC), where core 5G functions and user services are hosted. The architecture is divided into two network segments with varying trust levels: the provider network, run by the Mobile Network Operator (MNO), and the Data Network (DN). Key components facilitating the interaction between these segments are the User Plane Function (UPF) and the Network Exposure Function (NEF). The UPF manages the ingress and egress of user equipment traffic, while the NEF provides API-based access to 5G core information for MEC applications.

The second important part of the architecture is the "Edge Data Center" (Edge DC) which is in physical proximity to the RAN and the UE, thus, having advantages in terms of latency, data throughput or privacy compared to the Cloud DC. Services or Applications that require the properties of the MEC are deployed in the Edge DC.

4.1.3.2 Service Platform Architecture

As introduced with the network architecture, the service platform architecture consists of a trusted provider network and the DN, as well as a Cloud DC and an Edge DC.

Illustrated in Figure 4, the service platform architecture highlights the shared infrastructure model based on common cloud software and infrastructure components like Docker, Kubernetes, or KVM. This architecture spans across network, computing, and storage domains, all shared by services in a potentially multi-tenant environment. This shared model necessitates robust security measures such as TLS for data protection and strict capacity limits to curtail any malicious overuse of resources.

4.2 Focus for Risk Analysis

The risk analysis for MEC in 5G systems will focus on the security aspects of data transport, service communication, and shared platform usage. The primary scope includes detailed analysis of data integrity, authenticity, and encryption during transmission, encompassing communication between User Equipment (UE) and service instances, among service instances, and between DCs. The analysis also covers the utilization, allocation, and distribution of network and service platform resources, evaluating potential disruptions to critical MEC services due to malicious resource exploitation.

The secondary scope includes CPU and storage security, which are assumed to be secure and enforced by host systems and virtualization software. This area will be briefly described but not analyzed in detail. The ETSI MEC/3GPP 5GC API provided by the NEF, still in development and lacking standardized implementation, also falls under the secondary scope.

Out-of-scope elements, which will not be analyzed, include security weaknesses by system administrators, system provider companies, and software platforms such as containers and the Linux operating system.

These elements are considered as trusted. In this context, "trust" of system administrators refers to their loyalty to the company, i.e., that they do not willfully cause damage (e.g., through deliberate misconfigurations).

This division into primary, secondary, and out-of-scope categories should ensure a comprehensive and focused risk analysis, providing a solid foundation for understanding the technical intricacies and interdependencies within MEC in 5G systems.

Main references used for system architecture diagrams: [TS23501], [TS29501], [MEC003], [3GPPEC], [SYSARCH].

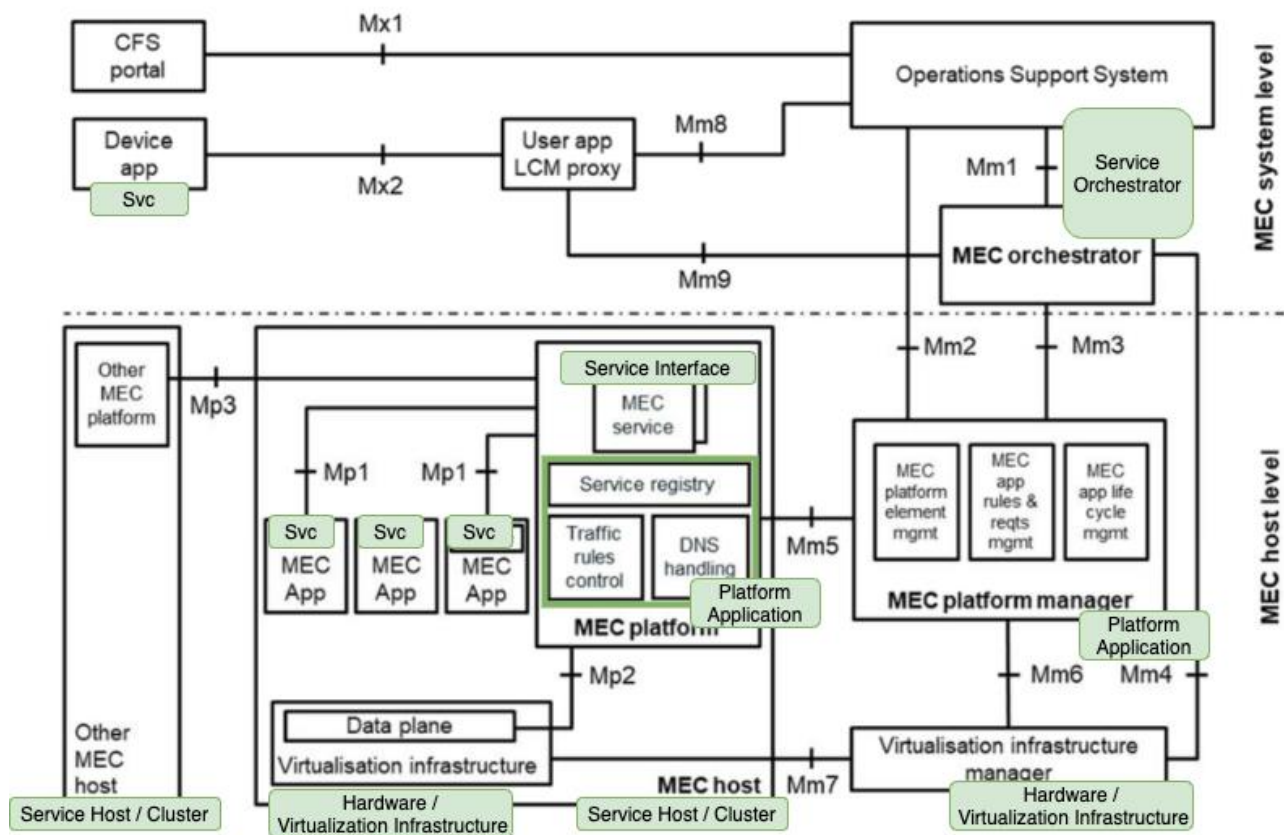


Figure 2: Mapping between ETSI GS MEC 003 V3.2.1 (2024-04) and the terminology (green) of the present report

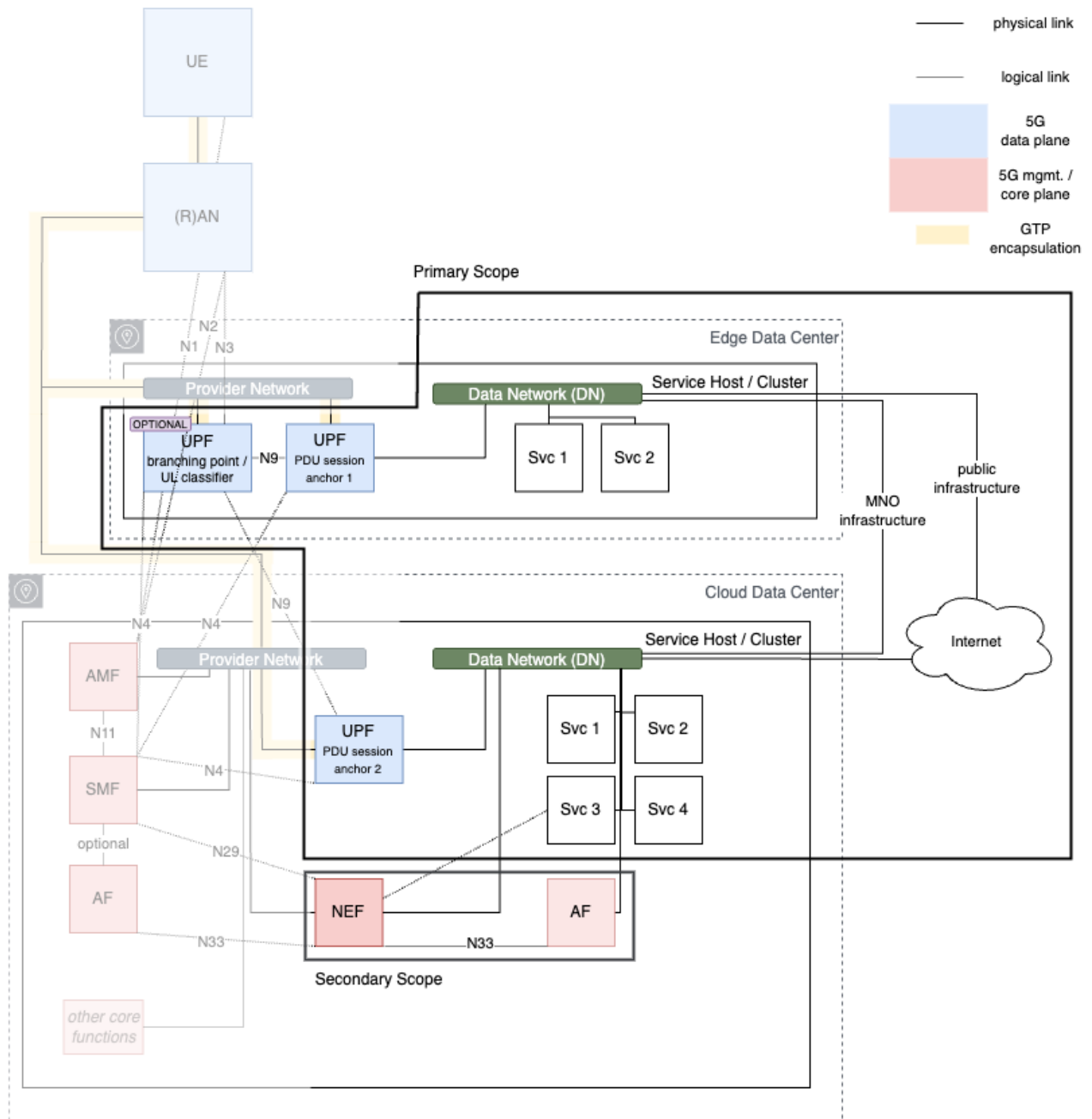


Figure 3: Network architecture

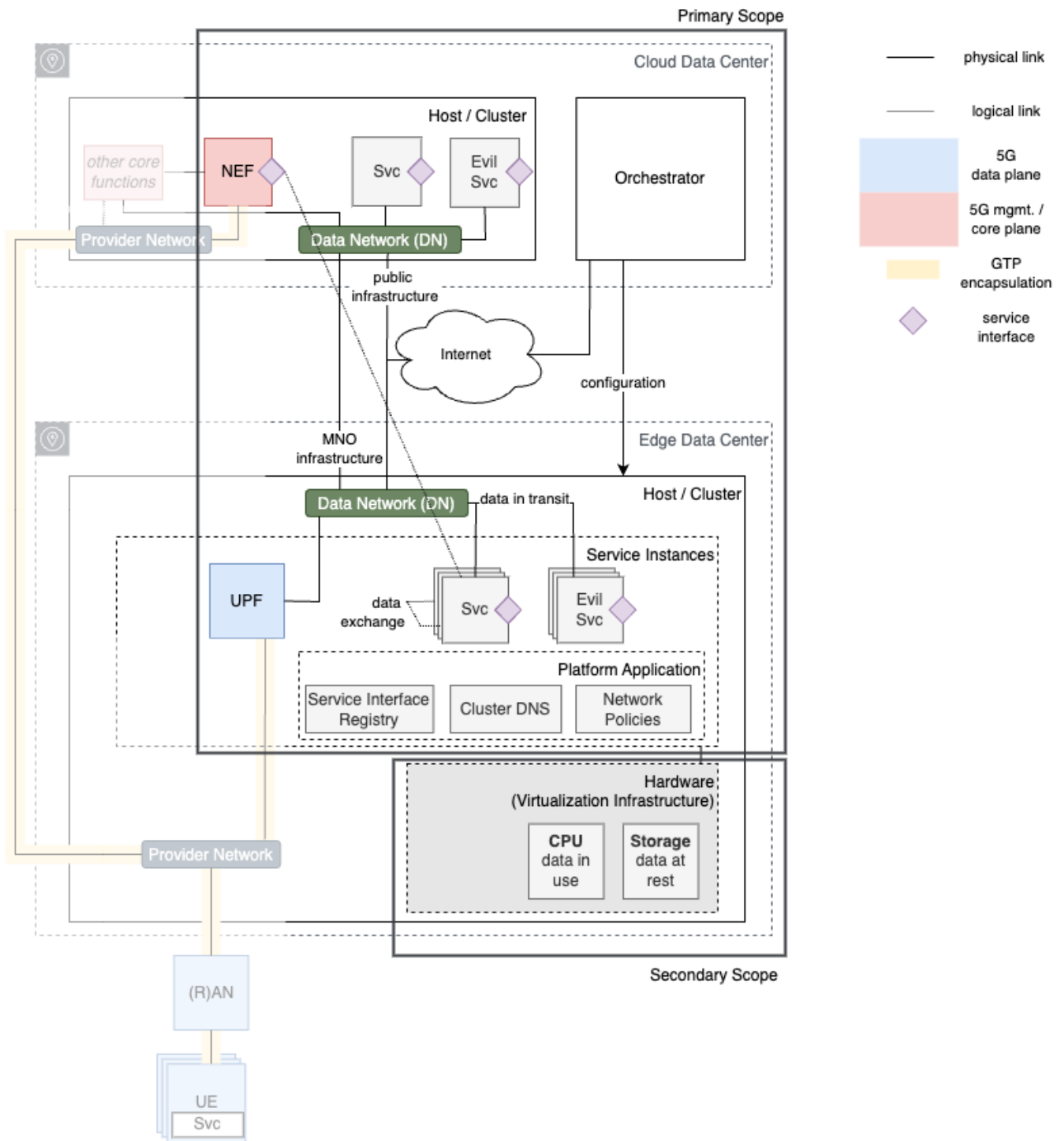


Figure 4: Service platform architecture

5 Risk Scenarios

This chapter provides the current view on a range of risk scenarios. It deals with an estimation of probabilities that each of this risk scenarios could become reality. These estimations are based on detailed discussions of potential scenarios of weaknesses or attacks. These discussions conclude with an assessment of the probability of occurrence.

The following table shows if MEC is a relevant sub system of each risk scenario. The risk scenarios themselves are further described in the framework document.

This risk analysis focuses on technical risks and risk scenarios. Covering political aspects is out of scope. Not taken into account are risk scenarios related to the question, if the manufacturer of the equipment and the supply chain is trustworthy. On this topic exist various publications for IT systems in general [ENI]. A second risk not covered by this analysis is the not standard-conform implementation of components. This analysis assumes vendors comply with 3GPP and other standards as long as a standard for the functionality exists.

Table 3: Relevant Risk Scenarios for the MEC Subsystem

Perspective	Section	Risk Scenario	MEC	CIA
Public MNO Perspective	5.2.1	Emergency Call (TKG § 164)		A
	5.2.2	Cell Broadcast (TKG § 164a)		A
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	x	C
	5.2.4	Emergency Preparedness (TKG § 184)		A
	5.2.5	Regional Network Failure		A
	5.2.6	Multi-Regional Network Failure		A
	5.2.7	Failure of Mobile Service	x	A
	5.2.8	Fraud		I
	5.2.9	Data Leak		C
User Perspective	5.3.1	Urgent Telephone Calls Not Possible		A
	5.3.2	Authenticity of the Receiver of a Call		I
	5.3.3	Integrity of Transmitted Data	x	I
	5.3.4	User's Identity	x	C
	5.3.5	Eavesdropping of a Communication	x	C
	5.3.6	Slowing Down/Restricting the Connection	x	A
	5.3.7	Connection Failure		A
	5.3.8	Tapping or Altering of the Geographical Position	x	C
Private 5G Network Operator Perspective	5.4.1	Network Failure	x	A
	5.4.2	Degraded Services/Connections	x	A
	5.4.3	Information Disclosure	x	C
	5.4.4	User Identities (Authenticity/ Integrity)	x	I

5.1 Attack Paths and Risk Sources

For each risk scenario a couple of different attack paths and risk sources will be discussed within this document. The result of the discussion will be a probability level for the impact defined in the respective risk scenario.

The term attack path refers to a sequence of steps or actions that an attacker can take to exploit vulnerabilities in a system or network, leading to security issues. It outlines the route or method an attacker follows to achieve his goals, often involving a series of interconnected actions that exploit weaknesses in a system's defences.

A risk source is defined as the origin or cause of a potential risk or threat. A risk source is identified as the factor or element that can lead to vulnerabilities or hazards.

As many attack paths or sub-sequences of attack paths and risk sources are similar for most of the risk scenarios, each attack path and risk source will be introduced from a MEC perspective in this chapter.

In general, an attacker with legitimate access to MNO's assets that shall be protected is called internal attacker. An internal attacker could have access to physical or cryptographic keys, to buildings or rooms or to networks or systems.

5.1.1 Misconfiguration (AP1)

Misconfiguration as a possible attack path concerns all configurable parts of a system. A legitimate configuration can either be deployed by employees of the network operator or the manufacturer or by configuration tools. The cause of misconfiguration can be either human error thus happening unintentionally, or it can also be intentional as part of an attack.

Misconfigurations may persist undetected if the impacted functionality was not used regularly or if configuration or security impacts of that functionality were not monitored.

5.1.1.1 Confidentiality (AP1-C)

(1) Privilege Abuse

Privilege abuse refers to an attack path from within the system. The root cause of most privilege abuse attack vectors is caused by not following the "principal of least privilege". On a user level, a possible AP occurs by attributing users higher than necessary privileges, allowing a malicious user to access system components or configuration of the system he is not intended to [SSDP]. In another AP in the context of MEC software, misconfigured privileges can be exploited by malicious software to read or write data or execute additional software on a system also affecting all pillars of security [ESEC] [SSDP].

In general, privilege abuse opens possibilities for further attacks on MEC components potentially affecting confidentiality, integrity, and availability. However, the focus within this section is on confidentiality.

(2) Service Identity Spoofing

Service identity spoofing can be a result of misconfigured authentication resulting in improper verification of identity. If no adequate API authentication is enforced by configuration, malicious services could present their APIs to users or services. Misconfiguration in DNS may lead to redirection of requests to malicious services. This can lead to unintentionally sharing data with malicious services breaking requirements on confidentiality and additionally on integrity and availability [ESEC] [BSI_Threat] [ENISA].

Probability Assessment

Unintentional or intentional misconfigurations pose a reasonable probability as reported in literature. Therefore, the probability of a successful attack on confidentiality due to misconfiguration is evaluated as **MEDIUM**.

5.1.1.2 Integrity (AP1-I)

(1) Insecure Protocol Configuration

Using default values for configuration parameters can lead to insecure protocol configuration as old or outdated security algorithms or protocols may be selected which can lead to an insecure data transfer. Additionally, disabling mechanism in protocols, such as certificate validation in a TLS handshake can lead to spoofed identities (see Service Identity Spoofing) [OWASPTLS]. Misconfigured logging systems which leave issues with the protocol or its authentication mechanisms undetected give an attacker the chance to remain undetected [MSHAF].

Probability Assessment

The integrity of MEC systems can be affected by misconfiguration. Only one specific attack path has been identified which results, due to its severity, to a **MEDIUM** probability.

5.1.1.3 Availability (AP1-A)

(1) Insufficient DoS Resilience

Denial of Service (DoS) attacks can occur if the MEC service is not properly configured to handle high volumes of requests or if rate-limiting, load balancing, and failover mechanisms are not adequately implemented. Such misconfigurations can result in service outages, degraded performance, and unavailability of MEC services. This especially affects applications that rely on low latency and high availability, leading to significant operational disruptions [ESEC] [SSDP].

(2) Insufficient Resource Planning (e.g., consideration of load peaks)

Within the MEC environment, insufficient resource planning can result in the system being unable to handle load peaks effectively. This can occur if resource allocation, scaling policies, and capacity planning are not properly configured to address varying demand levels. This can lead to system overloads, degraded service quality, and potential outages during peak load times. This can severely affect user experience and the reliability of applications that require consistent performance [RELSVC] [SSDP].

(3) Insufficient Network Separation (CUPS)

Insufficient network separation can occur if the control and user plane traffic are not properly isolated, potentially allowing unauthorized access or interference between different network functions. This can result in security vulnerabilities where malicious actors could exploit the lack of isolation to gain unauthorized access to sensitive control plane functions or disrupt user plane traffic. This can compromise the integrity, confidentiality, and availability of the 5G/MEC network services [ESEC] [C5] [SSDP].

Probability Assessment

The probability of a misconfiguration impacting the availability of a MEC system is evaluated as **HIGH**. This is, unfortunately, a very common root cause making systems unavailable. Recent incidents, i.e., the CrowdStrike incident [CROWDSTRIKE24], make it clear that sufficient testing or automatic rollback mechanisms are often not implemented and increase the probability of unavailability.

5.1.2 Insufficient Access Controls (AP2)

Insufficient access control poses a significant risk for a MEC system as it enables attacks potentially affecting all security goals. Additionally, due to the critical nature of access control, insufficient mechanisms can enable attacks on underlying subsystems.

5.1.2.1 Confidentiality (AP2-C)

With respect to confidentiality within a MEC system, the following main attack paths have been identified that result from insufficient access control.

(1) Malware

Malware can occur and infiltrate the system if user permissions are not properly managed, or if there are inadequate security measures to prevent unauthorized access and software installations. The presence of malware on the MEC platform can compromise the confidentiality, integrity, and availability of services. Malware can exfiltrate sensitive data, disrupt service operations, and provide a foothold for further attacks within the MEC environment [CYSEC] [SSDP].

(2) Service Identity Spoofing

Insufficient access control can lead to service identity spoofing as evil services may be deployed to a MEC platform if proper authentication is not implemented. Service calls will then be answered by evil services as no verification is in place. All data transmitted to an evil service experience the risk of potentially breaking all security goals, confidentiality, integrity and availability [ESEC] [BSI_Threat] [ENISA].

(3) Hardware Attacks Through Insufficient Access Control and Insufficient Hardening of the System

Insufficient hardening of the system opens up the possibility for attackers to gain access to confidential data and possibly access to private keys when physical access to the MEC component is possible [VfS], [E], [SSD]. An attacker who gains physical access to MEC hardware can perform hardware related attacks. Short interruptions or voltage spikes in the supply voltage can cause parts of a computer system, such as security components, to enter an undefined state. As part of the »faultTPM attack« [AMD_TPM], security researchers gained access to a TPM (Trusted Protection Module) from AMD by manipulating a voltage regulator, in addition to other measures. In the documented attack, the encryption of the hard disk could be broken in addition to other actions. Integrated circuits can be hardened against this type of attack through appropriate design measures [PGLITCH].

Probability Assessment

Overall, the probability of an attack path due to insufficient access control mitigating the confidentiality of a MEC system is assessed as **MEDIUM**. Especially the criticality of access control leads to this probability evaluation as insufficient access control opens possibilities for malware or service spoofing, as well as physical access to infrastructure of the MEC system. An attacker with physical access can perform advanced hardware related attacks to extract keys which may be used for data encryption.

5.1.2.2 Integrity (AP2-I)

(1) Network Function Identity Spoofing

Network function identity spoofing can occur if access control policies are weak or improperly enforced, enabling attackers to impersonate legitimate network functions and perform unauthorized network reconfigurations (e.g., injecting a malicious Application Function (AF) that is allowed to control UE ↔ UPF associations). Network function spoofing can lead to unauthorized network changes, service disruptions,

and potential security breaches. Attackers might redirect traffic, manipulate data, or degrade service performance, compromising the integrity and reliability of the 5G network and the MEC services [ESEC].

Probability Assessment

The probability of an attack path due to insufficient access control mitigating the integrity of a MEC system is assessed as **MEDIUM**. It is the criticality of access control that leads to this evaluation as it creates the possibility for network function spoofing such that the integrity of the MEC system cannot be guaranteed any more.

5.1.2.3 Availability (AP2-A)

(1) Insufficient Resource Capacity Limits

A malicious service may occupy a great number of resources of a MEC platform if resource capacity limits are not enforced by sufficient access control. Unauthenticated and unauthorized services can consequently attack the availability of the system. As a result, this leads to degradation of service response time and user experience. [CONSEC][SSDP]

(2) Credential / Identity Abuse

Stolen user credentials pose a serious threat if adequate multi factor authentication (MFA) is not in place. Using default credentials due to not enforcing password reset or checking for unallowed username and password combinations enabling rainbow list attacks (related to Misconfiguration), improper user management, long-lived credentials, no re-authentication of users, no limited session time may all lead to credential or identity abuse, potentially creating attack paths to the MEC. A successful attacker can exploit further vulnerabilities to affect confidentiality, integrity and availability of the system [E] [BSI_Threat] [ENISA].

(3) Dangers due to a lack of Physical Access Control

A lack of physical access protection can lead to the destruction or theft of MEC components, which primarily jeopardizes the availability (even if only locally). Subsequently, there is a risk that stored data or keys may be extracted (if the components are stolen) and attacks on other MEC or even the core network itself may also occur, for example by extracting the credentials for the N33 interface of the NEF [VfS] [E][SSDP].

Probability Assessment

The probability of an attack path due to insufficient access control mitigating the availability of a MEC system is assessed as **MEDIUM**. It is again the criticality of access control that leads to this evaluation. A MEC system that has insufficient access control could become unavailable due to maliciously acting users, occupying more resources than allowed. In general, a malicious actor that gains access could deactivate or render the system unavailable, either by remote or direct physical access.

5.1.3 Implementation Flaw (AP3)

Implementation flaws and poor quality of software or hardware can often affect confidentiality, integrity and availability at the same time. For example, TLS can be used for encryption and to secure integrity. So, a weakness in a TLS Implementation can compromise confidentiality and/or integrity. As a result, it is often not possible to exclusively assign an implementation weakness to the protection goal of confidentiality or integrity. Especially when implementation weaknesses enable follow-up attacks. In the following, vulnerabilities are assigned to the protection objective that is most frequently compromised in attacks. It should also be noted that no vulnerabilities in MEC implementations - listed on the ETSI website [ETSI_MEC_LIST] - are officially known at this time [CVE_Details]. The extent to which existing

vulnerabilities and exploits of protocols have an impact on MEC implementation is unclear, as it is unknown which libraries are utilized by each manufacturer. Interfaces that support communication with external entities and interfaces that are exposed to public networks (Internet) have a higher risk of being attacked than interfaces that are not accessible to an attacker. Exposed Interfaces/Reference Points are: Mp2 (MEC platform – UPF), Mx2 Management (UE, LCM, OSS, MEO), N33 IF (MEC-NEF), interface between RAN and MEC data plane (the actual decoupling of the data depends on the operator model).

5.1.3.1 Confidentiality (AP3-C)

Implementation errors or weaknesses pose potential risks to the confidentiality of data processed by a MEC system. These weaknesses can allow attackers to gain access to confidential and protected data.

(1) Malicious Applications

MEC relies on applications running on the MEC platform and on the UE that perform a variety of tasks. It is to be expected that attackers will attempt to get malicious applications onto the MEC platform and the UE in order to get access to confidential data. There are numerous examples of such APPs that remain undetected for a certain time in the Apple or Google App-Stores [HEISE_SPY], [MANDRAKE], [HEISE_TROJ] and infected mobile devices. The weakness here is inadequate testing during the release process and insufficient monitoring of the actions performed by an APP.

(2) Vulnerability or Bugs in Security Protocol Implementations

MEC uses different security protocols on different layers (network-layer, transport-layer) as well as token-based authentication and authorization (IPsec, TLS, OAuth2.0 and JWT). Vulnerabilities in the implementation can lead to attack vectors that result in encryption being broken or unauthorized access to confidential data.

A serious bug was the “Heartbleed” bug in the OpenSSL library. But, there are numerous weaknesses in token-based authentication and authorization [MICRO_OAUTH], [HACKER_STORM_0558] or vulnerabilities that have the potential to breach the confidentiality of data [CVE_Details], [OWASPTLS].

(3) Unsecure Key-Management (Key Generation, Key Storage, Key Exchange)

A weakness in the implementation of key management for encryption keys can affect the entire chain from key generation and key storage to key exchange or key sanitization. Weak key generator algorithms may result in weak cryptographic keys that are easy to break. An insecure key storage can lead to key extraction and weak key exchange procedures can facilitate man-in-the-middle attacks [FLUID] [TPM].

(4) Improperly Secured Exposure APIs

Exposure APIs that are not properly secured can lead to a data leakage, e.g. Location API. An attacker can possibly get information about location of devices [HACK].

(5) Weak or Missing Password Policy Implementation

A weak or missing implementation of password policy-checks (password structure, password change, hiding password display capabilities, consecutive failed login) can lead to unauthorized access. A weak password structure and/or a long period of password validity could lead to a successful brute force attack. Failure to block consecutive failed login attempts may lead to password guess [BSI_Threat] [ENISA].

Probability Assessment

The probability of a successful attack on confidentiality due to implementation flaws is **MEDIUM**.

Attacks on implementation vulnerabilities can lead to confidential data being accessible to unauthorized third parties. The use of open-source software can have both positive and negative impacts in this context. Due to its widespread use, vulnerabilities in open-source software are often discovered earlier than in closed-source software. It is important to apply available updates promptly after a vulnerability becomes

known. A potentially difficult-to-assess risk may arise from the APPs that can be deployed on the MEC platform. In this context, approval processes for the applications by the operator, depending on the rights required by the APP, can mitigate the risk of a threat. Additional measures such as antivirus scanners, APP monitoring, and reliable sandboxing can further reduce the risk of malicious APPs spying on the MEC. It is assumed that MEC platforms apply the security standards that are also used by reputable cloud service providers. If this is not the case, the risks can increase accordingly.

5.1.3.2 Integrity (AP3-I)

Implementation errors or weaknesses pose potential risks to the integrity of data processed by an MEC system. These weaknesses can allow attackers to manipulate the integrity of data. By exploiting these weaknesses, attackers may commit identity theft.

(1) Vulnerability or Bugs in Security Protocol Implementations

Weaknesses in the implementation of TLS, or token-based authentication can also result in attacks on the integrity and authenticity of data or users [CVE], [MICRO_OAUTH].

(2) Unsecure Key-Management (Key Generation, Key Storage, Key Exchange)

Weaknesses in the implementation of key management for signing keys affect the entire chain from key generation and key storage to key exchange or key sanitization. Weak key generator algorithms may result in weak cryptographic keys that are easy to break. Key extraction can result in signatures being forged or forged data being re-signed, thereby undermining the integrity and authenticity of digital signatures.

Probability Assessment

The probability of a successful attack on the integrity due to implementation flaws is considered to be **MEDIUM**. This applies to the integrity of the information and authenticity of the data exchanged between the 5G Core and the MEC platform. It is assumed that MEC platforms apply the security standards that are also used by reputable cloud service providers. If this is not the case, the risks can increase accordingly.

5.1.3.3 Availability (AP3-A)

Implementation errors or weaknesses pose potential risks to the availability of the MEC platform itself. The impact on the availability of the 5G network, on the other hand, depends on the stability and configuration of the 5G Core itself. The following are implementation weaknesses that enable attackers to weaken the availability of IT systems through corresponding attacks, as direct attacks on MEC platforms are currently not known or published.

(1) Insufficient Quality of Software (MEC Services and APPs)

MEC nodes can be opened to allow authorized participants to deploy applications/services to other users. However, poorly designed applications can create opportunities for attackers to invade the system and pose threats to the network applications running on the platform [NIH].

So, software of inadequate quality can be a target for attackers at any point. The higher the authorizations of a software, the greater the potential damage. The MEC APPs in particular can be targets for attackers, especially if they have an interface and communication to external entities (e.g. user equipment, Internet). Possible attacks could be: buffer overflows, code injection or opening backdoors and potential follow-up attacks like aiming at exploiting the virtualisation environments, and even denial of service attacks are possible [OWASP_M7], [BSI_Threat], [ENISA], [SSDP].

(2) Insufficient Input/Output Checking and Validation

Insufficient validation and sanitization of data from external sources, such as user inputs or network data, can introduce severe security vulnerabilities, e.g. code injection, command Injection, cross-site scripting (XSS) and buffer overflows. [OWASP_M4], [NCC], [JSON], [CWE_20]

(3) Weakness in Parser

XML-Parser: So-called XXE injections (XML External Entity attack) can be used to read files on the attacked system and, if accessible, interact with other internal or external systems and can also be used for DoS attacks. Other known exploits are XML Bomb and further exploits leading to heap overflow or deadlocks.

JSON-Parser: Even though JSON parsers are considered less susceptible to DoS attacks, there are still known vulnerabilities that can enable a DoS attack. A security gap exists if the JSON string can be manipulated in such a way that a class is instantiated during deserialization that can cause damage (denial of service, but also sensitive data exposure, data manipulation, etc.) [JSON].

(4) Weakness in the Deployment of MEC “Firmware”, Services and APPs.

Vulnerabilities or a lack of security checks during the deployment (distribution and installation) of software can lead to malicious software being installed on the system. CWE-829: Inclusion of Functionality from Untrusted Control Sphere, CWE-494: Download of Code Without Integrity Check, and CWE-502: Deserialization of Untrusted Data. Targeting the MEC by a poorly configured CI/CD pipeline and the ingest of malicious code could also be a threat [NCC], [CWE].

(5) Lack of Redundancy

The lack of redundancy in planning and implementation can lead to the MEC's services no longer being available if a system fails. The effects depend on the criticality of the offered services [VfS] [E].

Probability Assessment

The probability of a successful attack on the availability is considered to be **MEDIUM**.

The prerequisite is that the MEC platform itself is implemented according to current security standards and measures to secure and defend against attacks on availability are in place. A threat can arise from the possibility that third-party applications can be installed on the system, whose quality and stability cannot be completely verified. However, this does not affect the availability of the overall system.

5.1.4 Specification Weakness (AP4)

The current specification work in ETSI MEC and 5G is based on virtualization frameworks, to allow a strong separation of modules with a resilience to overload scenarios and software crashes. Their interaction is specified via communication protocols and interfaces. Security mechanisms are defined for authentication, encryption and integrity protection. For control of information passing the interface, a control of the reachable interfaces was introduced by 5G, namely the Policy Control Function (PCF). Also, the access to the 5G Core network is limited by the Network Exposure Function (NEF) regarding bandwidth in order to prevent overloading attacks. The secure encapsulation of an external part of software (APP) is rather important in contrary to a review of all APPs.

ETSI MEC is supposed to be the option used by PLMNs, because of the well evaluated specification and the ongoing specification work, especially concerning the quantum cryptography. In private networks, which are used in the industry, real time ability is one of the core features. Therefore, the use of company specific solutions with unknown specification is expected due to their market readiness. The APPs in an industrial scenario are restricted and quality tested. And there may be limitations to reach the WAN, to avoid industry espionage. The only way to assess the risks of such a proprietary MEC is to check its influence on 5G via its interfaces. And finally, a logging mechanism shall allow to monitor all errors and suspicious activity. Ongoing specification work can further improve the quality. Still, incompatibilities between versions and

fall back to lower versions are typical security issues. At present no documented specification weaknesses of the ETSI MEC could be found.

5.1.4.1 Confidentiality (AP4-C)

ETSI MEC utilizes 5G network slicing as a security feature. Weaknesses of slicing have been documented but will not be treated here. Following threats are seen:

(1) Misuse of the Location Interface of 5G to Track a Person.

By finding the approximate home and the approximate working place the person may be identified. The ability of an UE-APP to leak GPS-data is not considered here.

(2) Extraction of the Users Identity by Bypassing the Policy Control Function and Accessing the UDM Directly

If not prevented by specified mechanisms, MEC may have access to user data including the 5G-Key K for EAP-AKA and EAP-AKA'. This is a potential threat for the confidentiality of data. The protective mechanism is in the 5G-Core but will be discussed here. A similar attack may be done by MEC charging (billing) via 5G charging infrastructure as charging means assigning costs to an identity.

(3) Copy of Voice and Data routed by the UPF

A MEC could get access to the legal interception mechanisms to route a copy of some connections to a MEC APP or a third party. If the UPF and the MEC-virtualization uses misconfigured switches, IP-frames could be sent to more than one switch-port. Example: Debug mode of Ethernet switches. The lawful interception is an example of such a rerouting.

Probability Assessment

Policy control function and NEF can prevent access to the location interface provided by the AMF excluding (1). Both the UDM and SMF can be protected by the Policy Control Function excluding (2 and partially 3). The network administrator has to take care of a proper selection and configuration of hardware switches excluding (3). Virtualization typically uses switches with a good level of security excluding (3). Overall, the probability of successful attacks on the confidentiality is seen **LOW**.

5.1.4.2 Integrity (AP4-I)

Attacks to the Integrity of Data can take effect during the communication or inside the MEC-APPs. The latter is not covered by the ETSI MEC specification but is usually done with signatures on application level. So only the attack scenarios in communication are treated here.

(1) Man-in-the-Middle Attack to Communication Between MEC Modules

As a MEC system may be distributed over several servers, the communication between modules is routed via LAN or WAN and could be attacked by a man in the middle attack, manipulating data.

Probability Assessment

The ETSI MEC specification recommends usage of TLSv1.3 to protect both confidentiality and integrity of inter module communication. Mutual authentication is done by certificate chains coming from a trusted Certificate Authority. The probability of successful attacks on the integrity is **LOW**.

5.1.4.3 Availability (AP4-A)

Availability can be reduced by the following threats:

(1) Software bugs causing memory and calculation leaks.

A container will be restarted in higher frequency than the crashed container can be removed. This can also be caused by interface queries, which cause unrecoverable errors.

(2) MEC APP causing high calculation power needs

Spamming of a module with requests beyond the answer time may start a large number of containers thus using all the calculation power.

(3) MEC APPs overload the data throughput of the UPF

One or more MEC APPs may react to queries with large amounts of data, or a UE APP may query in high frequency. As the overall performance of the RAN is limited, other traffic is slowed down or stopped.

(4) Attack of configuration interfaces and misconfiguration

Misconfiguration can stop both MEC and 5G. E.g. granting very low resources to the APPs or stopping a relevant APP like MEC Service registry.

Probability Assessment

Threat (1) and (2) are typical scenarios, virtualization and orchestration can deal with. Logs and alerts allow transparency and debugging of the system. The system remains stable while an operator can take care of the erroneous APP. Threat (3) is a matter of system design and configuration. The bandwidth of connections can be configured so that resources for special services are reserved. Protection of configuration interfaces (4) is well specified. So, the probability of a successful attack on the availability is considered to be **LOW**.

5.1.5 Failure of Power Supply (AP5)

The multi-access edge host systems are positioned at the edge of the network. They are located close to the user and often in open, unsecured environments. Therefore, MEC servers can be less secure and more susceptible to physical attacks than servers hosted in data centers with proper physical protection. There is a higher risk of unauthorized access to the devices, which can lead to functional impairment or damage. This jeopardizes the availability and confidentiality of stored or transmitted data [SSDP].

Various factors can lead to problems with the power supply of edge systems. In addition to the classic technical failure of a power supply unit, which usually leads to the total failure of the respective functional components supplied, there is also the possibility of an intentional shutdown or destruction. A further fault can be triggered by an unforeseen failure in the power supply. In the internal power supply, individual phases can fail, e.g. due to overload. There is also the possibility of a regional failure at the grid operator. Various measures can be taken to harden the power supply of edge components in the field of edge computing. Redundantly designed power supply units, typically used inside server components, do not necessarily lead to a component failure in the event of a power supply unit defect. If these power supply units are connected to different phases, the redundant phase takes over the supply of the component in the event of a power supply segment failure. Uninterruptible power supplies can bridge short-term power failures. In the event of longer interruptions, emergency power generators supply the most important system components.

A power supply failure has various consequences. The decommissioning of the central server component in a distributed edge computing system due to a power supply failure usually leads to a complete failure of services. If less important peripheral components are affected, this results in a partial failure. This can degrade individual services.

There are differences between multi-access edge computing in a public 5G network and a private 5G network in the event of a power failure. In a public network, the computing resources of edge computing are distributed over a larger area but placed close to the application to reduce latency. Regional power failures therefore tend to lead to partial failures. If a few components fail, servers further away may be able to take over the tasks of the failed computers. In the private 5G network, larger computing capacities are installed in a small number of server cabinets. A failure due to a power interruption can quickly lead to a complete failure in this constellation. However, redundant structures and emergency systems such as uninterruptible power supplies or emergency power generators are easier and cheaper to install in a campus network.

5.1.5.1 Confidentiality (AP5-C)

During this analysis, no relevant attack path was found that would lead to a breach of confidentiality in the event of a power supply failure.

Probability Assessment

Based on current knowledge, it cannot be ruled out that attack paths exist or will exist. The risk of a breach of confidentiality is therefore considered to be **LOW**.

5.1.5.2 Integrity (AP5-I)

In the event of a power failure, it should be noted with regard to integrity that the supplied components usually fail spontaneously. This can happen during a storage process or data transfer. In this case, data can be corrupted or lost. Measures such as checksums can be used to detect this or correct errors. As a rule, the bigger problem is the failure of the respective component. If file systems are corrupted, this can lead to systems not booting up correctly when they are restarted (violation of system integrity).

Probability Assessment

Additional protective measures can make direct access to the power supply of edge components more difficult. When using the measures described above, such as redundant power supply units, uninterruptible power supplies and emergency power generators, the risk of data loss or data corruption in the event of a spontaneous power failure is **LOW**.

5.1.5.3 Availability (AP5-A)

Interruption of the power supply

The same mechanisms as in apply to this attack path. A power supply failure that cannot be bridged over time at one or many edge components leads to a total failure. This affects the availability of individual or all edge services for the duration of the failure.

Probability Assessment

Additional protective measures can make direct access to the power supply of edge components more difficult. If the measures described above are used, such as redundant power supply units, uninterruptible power supplies and emergency power generators, the risk of data loss or data corruption in the event of a spontaneous power failure is **LOW**.

5.1.6 Failure of Hardware Components or Unavailability of Spare Parts (AP6)

Due to the reduced physical protection of servers (e.g., industrial PCs) in the edge area, there is a higher risk that unauthorized individuals may gain access to an edge component and manipulate it. Furthermore, the devices are often operated in locations that, due to the necessary latency, are close to data collection points in order to process data in real time. These locations are typically not climate-controlled compared to operating in a large data center. Therefore, these systems may be exposed to aggressive environmental conditions that can lead to technical problems over time. This is a natural degradation of the hardware or possible quality problems during the manufacturing process. Typical sources of error are failure of the power supply units due to possible overheating, power surges or voltage peaks. The CPU can overheat and age faster due to dust obstructing the airflow, clogged fans and heat sinks. Similar mechanisms damage the memory used. Fixed storage media and hard disks can be damaged and fail due to wear and tear over time or due to writing processes that have a physically destructive effect on the storage medium.

Various measures can be taken to harden edge components in edge computing against the effects of hardware failures. Distributed computing resources and spare components achieve greater resilience against failures. In a public network, the computing resources of edge computing are distributed over a larger area but placed close to the application to reduce latency. Limited hardware failures at a single location are therefore more likely to lead to partial failures. If individual components fail, servers further away may be able to take over the tasks of the failed computers. In a private network, computing capacities are installed in a small number of server cabinets. In this constellation, a device failure quickly leads to a complete failure of important local services if specially planned replacement components cannot take over the tasks. It should also be noted that embedded systems are often used in this area, which have a lower MTBF than high-quality server components.

5.1.6.1 Confidentiality (AP6-C)

During this analysis, no relevant attack path could be found that would lead to a breach of confidentiality in the event of a hardware failure of computer components of multi-access edge computing. If the analysis is carried out at the level of functional components in hardware modules such as a server processor or a Trusted Protection Module, it cannot be ruled out that a hardware failure occurring at this level could soften security barriers implemented by design or lead to security-relevant malfunctions.

Probability Assessment

The probability of errors at hardware level with a serious impact on security is low. Possible errors occur sporadically and cannot be deliberately provoked. The risk of a breach of confidentiality due to a power failure is therefore **LOW**.

5.1.6.2 Integrity (AP6-I)

In the event of a hardware failure, it should be noted about integrity that the affected components usually fail spontaneously or exhibit unexpected behavior. This can happen during a storage process or the transfer of data. In this case, data can be corrupted or lost. Measures such as checksums can be used to detect this or correct errors. As a rule, the bigger problem is the failure of the respective component. If file systems are corrupted in the process, this can lead to systems not booting up correctly when they are put back into operation after a repair (violation of system integrity). Security components such as TPMs should be designed in such a way that data is not corrupted or disclosed in the event of a hardware failure. However, incorrect behavior due to design errors cannot be ruled out. For example, attackers used side-channel attacks to gain access to TPM modules from AMD [AMD_TPM]. Design errors could be the reason for this.

Probability Assessment

The use of redundant structures in the form of spare components or server components that are further away from the location, standard measures such as checksums and the use of special robust hardware reduces the risk of a hardware error in an edge component jeopardizing the integrity of the edge system and edge services. If these measures are used, the risk of data loss or data corruption due to hardware faults is **LOW**.

5.1.6.3 Availability (AP6-A)

(1) Interference due to Environmental Influences

Computer systems at the edge are often operated in non-climate-controlled environments, or the physical protection to access the climate control is not sufficient. Environmental factors can thus impair correct operation. Typical effects include a reduction in performance or a shutdown of the component. Critical influences can be general ageing, insufficient cooling, extreme humidity, water ingress and corrosion, or the outbreak of fire. As a result of a hardware failure, the affected component can no longer maintain regular operation. Depending on the importance of the component, this leads to a partial or complete failure of the Multi-Access Edge Computing system.

The introduction of redundant structures in the form of spare components reduces the risk of a hardware failure in one edge component leading to a total failure of all edge services. Various manufacturers offer so-called edge servers that are designed in such a way that they can be operated reliably in non-air-conditioned environments over a long period of time. In public networks, there are often many computing resources that can take over the tasks of a failed component in the event of a hardware failure. In a private 5G network, edge services do not use servers alone. Servers often work together with industrial PCs, industrial controllers and embedded systems such as a Raspberry PI. These components generally have no redundancy but are replaced in the event of a hardware failure with replacement hardware held in stock. In systems where a high level of operational reliability is required, redundant system components should be used as spares.

Probability Assessment

The measures described, such as the use of redundant systems and the use of special hardware offered for use at the edge, reduce the risk of hardware failure or mitigate the consequences. If these measures are used, the risk of system failures due to hardware faults is estimated to be **LOW**.

5.2 Public Mobile Network Operator (MNO) Perspective

This section discusses the risk scenarios related to the MNOs perspective.

For each subsection following, a short evaluation of the relevance of the risk scenarios in the MEC is discussed and the probability that potential attack paths could be exploited is derived.

5.2.1 Emergency Call (TKG § 164)

Multi-access edge computing works on its own servers outside the 5G core network and has no access to the system for emergency calls. An attacker will therefore not be able to carry out an attack on the emergency call functionality of the 5G network from the MEC.

Multi-access edge computing is therefore not considered as a relevant attack domain for the “Emergency Call” risk scenario.

5.2.2 Cell Broadcast (TKG § 164a)

Multi-access edge computing works on its own servers outside the 5G core network and does not have access to the network to trigger a cell broadcast. An attacker will therefore not be able to carry out an attack on the cell broadcast capability of a 5G network from the MEC.

Multi-access edge computing is therefore not considered as a relevant attack domain for the “Cell Broadcast” risk scenario.

5.2.3 Telecommunications Secrecy (TTDSG § 3)

The protection of telecommunications secrecy in accordance with "TDDDG § 3 Confidentiality of Communications" is just as crucial for MEC as it is in the 5G network. It must be ensured that metadata and user data processed in MEC services are only used for the purpose specified in TDDDG § 3. It must also be ensured that this data is not passed on to unauthorized persons or intercepted by unauthorized persons. So, this risk scenario is relevant to MEC.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The probability of the risk of breaching telecommunications secrecy due to misconfigurations within the public MNO perspective is MEDIUM. Privilege abuse, resulting from not adhering to the principle of least privilege, allows malicious users access, enabling further attacks on MEC components and compromising confidentiality of user data. Service identity spoofing, due to misconfigured authentication and inadequate API verification, can redirect users to malicious services, breaching telecommunication secrecy by leaking user data. Insecure protocol configurations, often from using default values and outdated protocols, lead to insecure data transfers and potential identity spoofing. If logging is not appropriately managed or misconfigured, vulnerabilities and attackers can remain undetected [ESEC, SSDP, MSHAF, OWASPTLS]. As misconfiguration is likely to happen especially by accident (e.g., administrator is not aware of side-effects of a particular configuration), this attack path collectively threatens telecommunication secrecy by exposing sensitive data and disrupting service integrity and availability. Therefore, the probability of this scenario is rated as MEDIUM.	Medium
Insufficient Access Controls	Confidentiality (AP2-C)	The probability of the risk of breaching telecommunications secrecy due to insufficient access control within the public MNO perspective is MEDIUM. Insufficient access control can lead to significant risks such as malware infections and service identity spoofing. Malware can infiltrate the MEC platform, compromising confidentiality, integrity, and availability by exfiltrating sensitive data, disrupting services, and enabling further attacks. Similarly, lack of proper authentication can allow evil services to impersonate legitimate ones, intercepting and corrupting data transmissions. For a public MNO, this poses a serious risk to telecommunication secrecy, as sensitive communications could be intercepted and manipulated [CYSEC, SSDP]. As there is a great awareness about this obvious attack path, operators usually take great care to implement appropriate security measures. Nevertheless, the probability of this risk is evaluated as MEDIUM due to the root cause being insufficient access controls and the broad possibilities of inadequate security measures being in place. This can lead to a severe breach of telecommunication secrecy, undermining user trust and regulatory compliance.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	The probability that the confidentiality of information will be breached due to implementation errors or weaknesses is estimated to be MEDIUM. The classification in level MEDIUM is based on the ongoing reporting of implementation flaws in software that allow unauthorized access to data. Exemplary vulnerabilities are the Heartbleed vulnerability in OpenSSL (TLS), but also flaws in the protection of private keys and implementation errors in the generation of authentication and authorization tokens [MICRO_OAUTH], [Storm-558-Attack], [OAUTH], [CVE_OAUT2_0]. In addition, when using certificate-based authentication, validation errors during certificate generation or certificate verification are known that increases the risk of unauthorized access [Fortinet_TLS_Cert], [DigiCert].	Medium
Specification Weakness	Confidentiality (AP4-C)	Currently, no weaknesses in the ETSI MEC specifications have been published. MECs asymmetric key material is included in the APPs data space and the separation of APPs, authentication and authorization mechanisms, encryption and integrity protection algorithms are state of the art. Both unwanted access from one APP to another APPs data space and access to other 5G communication are prevented by these mechanisms. Access to key material of 5G is prevented. Therefore, the probability of the risk breaking telecommunication secrecy by exploiting specification weaknesses is LOW.	Low
Failure of Power Supply	Confidentiality (AP5-C)	As part of the analysis, no relevant attack path could be found that would jeopardize the confidentiality of telecommunications secrecy in the event of a power supply failure. As it cannot be completely ruled out that attack paths exist or will exist, the probability of the risk of a breach of confidentiality is assessed as LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	As part of the risk analysis, no relevant attack path could be identified that would lead to a breach of confidentiality in the event of a hardware failure of complete computer components. If parts of components such as the server processor or a Trusted Protection Module fail, it cannot be ruled out that the hardware failure that occurs will soften security barriers introduced by design or lead to security-relevant malfunctions. Special design measures should be taken to avoid this type of error. It can therefore be assumed that they occur extremely rarely and unpredictably. It is difficult for potential attackers to foresee and exploit them. For this reason, the probability of the risk of a breach of confidentiality is assessed as LOW.	Low

5.2.4 Emergency Preparedness (TKG § 184)

Multi-access edge computing does not usually have access to the administrative functions of the mobile network. Therefore, administrative tasks such as prioritizing users and services in an emergency are not performed via MEC services.

Multi-access edge computing is therefore not considered as a relevant attack domain for the "Emergency Preparedness" risk scenario.

5.2.5 Regional Network Failure

Multi-access edge computing generally does not have access to the radio access network in order to attack the availability of base stations within the RAN and thus provoke a regional network error. The MEC system only has access to the 5G core network via the Network Exposure Function, which acts as a security gateway. The possibilities of MEC access to the 5G core network and the RAN are defined in the 5G standard. The specific access rights of MEC functions are regulated by policies.

Multi-access Edge Computing is therefore not considered as a relevant attack domain for the "Regional Network Failure" risk scenario.

5.2.6 Multi-Regional Network Failure

Multi-access edge computing usually does not have access to the radio access network or the 5G core network in order to attack the availability of various base stations or 5G Core components and thus provoke a multi-regional network error. The MEC system only has access to the 5G core network via the Network Exposure Function, which acts as a security gateway. The possibilities of MEC access to the 5G core network and the RAN are defined in the 5G standard. The specific access rights of MEC functions are regulated by policies.

Multi-access edge computing is therefore not considered as a relevant attack domain for the "Multi-Regional Network Failure" risk scenario.

5.2.7 Failure of Mobile Service

Failures of MEC services and applications can result in certain functions no longer being available to users. If and how operators provide redundancy is unknown during this analysis. Similarly, the criticality and therefore the future applications can hardly be assessed. This must be determined through Service Level Agreements. It is also important to distinguish between the MEC service platform itself and the potentially offered MEC services or applications. It is assumed that higher quality standards are applied and implemented for the MEC service platform than for the MEC applications themselves, which are created by third-party providers. So, the MEC system is considered as a relevant attack domain for the "Failure of Mobile Service" risk scenario.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The probability that a mobile service in the MEC system fails due to misconfiguration within the public MNO perspective is HIGH. Misconfiguration issues in a public MNO's MEC environment, such as insufficient resource planning, inadequate DOS resilience, poor network separation, or errors in network configuration (DNS, routing tables) can impact the risk of "Failure of Mobile Service". Insufficient resource planning, DoS resilience or a wrong network configuration can lead to service outages on a MEC, and degraded performance during high traffic periods. This affects especially applications that depend on low latency and high availability. Poorly configured network separation together with the developers' assumption that management interfaces and traffic on the control plane are secured by default (but in fact they are often not secured by default) can create security vulnerabilities, allowing unauthorized access and interference, potentially disrupting service integrity and availability. The root cause of these issues is inadequate configuration or planning. The probability of this risk is considered HIGH in regard to the complexity of distributed systems, and because misconfiguration is a very likely root cause for the failure of mobile services [ENISA, ESEC], in this case specifically for MEC services.	High
Insufficient Access Controls	Availability (AP2-A)	The probability that a mobile service in the MEC system fails due to insufficient access control within the public MNO perspective is MEDIUM. Insufficient access control, such as improper credential management or lack of resource capacity limits, can lead to malicious services overloading a MEC platform, causing MEC service degradation. Additionally, weak authentication practices can result in credential abuse, allowing attackers to set malicious configurations, degrading the MEC mobile service [ENISA, SSDP]. For a public MNO, these issues can significantly increase the risk of mobile service failure, disrupting user experience and service availability. Insufficient access control can lead to a variety of possible attacks for network degradation as mentioned before. Therefore, the probability of this risk is MEDIUM. This is due to the critical nature and complexity of proper authentication and resource management.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	The probability that an error or weakness in the implementation will cause a failure of a MEC service or application is generally classified as MEDIUM. Currently, there is no reliable information available regarding the practical reliability of Multi-Access Edge Computing (MEC) and distributed MEC-APPs and MEC-Services in 5G networks. The classification into level MEDIUM for MEC applications and services has been chosen based on the assumption that MEC applications and services are tested in different depths within the release processes. This assessment is a result of the varying software quality of applications that are made available to users via the APP stores for Android and iOS devices. As an extremely negative example, it was possible to distribute malware through the APP stores until detection occurred [MAL], [Android_Malware], [iOS_Malware]. Similar issues have been an ongoing concern regarding MEC APPs and should be anticipated in the future.	Medium
Specification Weakness	Availability (AP4-A)	Currently, no weaknesses in the ETSI MEC specifications have been published. The mechanisms to control a flawed, attacking or attacked APP are sufficient to encapsulate the issue in a defined amount of memory and calculation power. Only the services of the APP causing the issue will not be available. Therefore, the probability that an error or weakness in the specification will cause a failure of a mobile service in a MEC system is generally classified as LOW.	Low
Failure of Power Supply	Availability (AP5-A)	In addition to the risk of a general power failure, the less protected area in the edge makes it easier to attack the power supply. A failure of the supply voltage at a central or many edge components that cannot be bridged over time leads to a total failure. This affects the availability of individual or all edge services for the duration of the outage. Additional protective measures can make direct access to the power supply of edge components more difficult. If these and other measures such as redundant power supply units, uninterruptible power supplies and emergency power generators are used, the risk of mobile services failing in the event of a power failure is LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Edge components often operate in non-air-conditioned and less protected environments. In these locations, there is a risk of environmental influences or the effects of force disrupting the operation of the hardware or destroying the component. The use of redundant structures in the form of spare components reduces the risk that a hardware fault in an edge component will lead to a total failure of all edge services. In public networks, there are often a large number of computing resources that can take over the tasks of a failed component in the event of a hardware failure. If measures such as physical protection of the hardware and the use of redundant systems are implemented, the risk of system failures due to hardware faults is LOW.	Low

5.2.8 Fraud

This scenario addresses fraud on billing services in a public 5G network. Multi-access edge computing does not use any billing systems of the 5G network.

Multi-access edge computing is therefore not considered as a relevant attack domain for the “Fraud” risk scenario.

5.2.9 Subscriber-Data Leak

Multi-Access Edge Computing does not have direct access to the user database (UDM) within the public 5G network. Therefore, this confidential data cannot be leaked within the MEC.

Multi-access edge computing is therefore not considered as a relevant attack domain for the “Subscriber Data Leak” risk scenario.

5.3 User Perspective

This section discusses different risk scenarios from the perspective of a user of the public mobile network. Private 5G networks are not in scope of this section. Since some of the risk scenarios one might typically expect here (e.g. emergency calls) are already described in the previous paragraphs, they won't be mentioned again in this section.

5.3.1 Urgent Telephone Calls Not Possible

Multi-access edge computing cannot influence telephone calls.

Multi-access edge computing is therefore not considered as a relevant attack domain for the “Urgent Telephone Calls Not Possible” risk scenario.

5.3.2 Authenticity of the Receiver of a Call

Multi-Access Edge Computing is not involved in the routing of telephone calls and cannot influence telephone calls.

Multi-access edge computing is therefore not considered as a relevant attack domain for the “Authenticity of the Receiver of a Call” risk scenario.

5.3.3 Integrity of Transmitted Data

There are vulnerabilities that can lead to a breach of the integrity of user data that is transmitted to and from the multi-access edge computing system.

Multi-access edge computing is therefore considered as a relevant source of risk for the authenticity of transmitted data.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	The probability of breaching integrity of transmitted data due to misconfiguration within the user's perspective is LOW. Misconfiguration in a 5G MEC system can impact the integrity of transmitted data. Insecure protocol configurations can allow attackers to intercept and alter data during transmission [OWASPTLS]. Service identity spoofing, such as DNS attacks, can mislead users into connecting to malicious services, leading to data tampering [DNSPOISON]. Privilege abuse within the system can lead to unauthorized modifications of data [CONSEC]. For a user accessing MEC services, these vulnerabilities increase the risk of data being altered or corrupted during transmission. However, it is assumed that services are implemented according to well-known best practices and nowadays the communication between internet services (e.g., UE APP <-> MEC APP) widely employs transport encryption using TLS, and DNSSEC for DNS protection for critical services. Those protocols implicitly verify the integrity of submitted data. Consequently, unless TLS/DNSSEC are misconfigured or missing, the probability of such an attack vector being exploited is evaluated as LOW, because this is handled on higher layers by the protocols being used. A misconfiguration, e.g., of TLS, is normally detected and handled by client libraries (e.g., known from certificate warnings when using HTTPS in web browser).	Low
Insufficient Access Controls	Integrity (AP2-I)	The probability of breaching integrity of transmitted data due to insufficient access control within the user's perspective is MEDIUM. Insufficient access control can lead to injection of malware and service identity spoofing on the MEC platform, both of which severely affect the integrity of transmitted data. Malware, stemming from poor user permission management, can exfiltrate data and disrupt services. Service identity spoofing allows malicious services to impersonate legitimate ones, leading to unauthorized data interception, consequently, data integrity cannot be guaranteed. From a user perspective, these vulnerabilities significantly compromise the integrity of transmitted data [CYSEC, SSDP, CONSEC]. The probability of this risk is MEDIUM due to the ease of exploiting weak access controls. The root cause is inadequate security measures and lack of proper authentication mechanisms.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	The probability that the integrity of transmitted data will be breached due to implementation errors or weaknesses is estimated to be MEDIUM. Even if there are no known CVEs or exploits [CVE_Details] for ETSI-compliant implementations [ETSI_MEC_LIST], there are known exploits for protocol implementations and token-based access methods that are used within the MEC system, which lead to an assessment in level MEDIUM. The protection of user data integrity between UE client APP and MEC server APP is carried out up to the UPF through the 3GPP protocols, if data protection is enabled. On the UPF, the data can be unencrypted and unprotected, but this primarily concerns the 5G Core Network. The decoupling of data from the UPF to the MEC APPs and the protective measures on this communication path are not part of the 3GPP and ETSI specifications and are implementation specific. In general, it is recommended to secure user data using TLS (encryption and integrity) and for authorization OAuth2 or JWT [MEC_Sec]. Thus, the integrity of the data is subject to the risks and flaws of the libraries that are used. According to TS 23.558 [TS 23.558], the protection of interfaces for application management (download, instantiation, execution) is carried out via TLS and token-based access procedures. Thus, ensuring the end-to-end integrity of the transferred data is the responsibility of MEC applications (consisting for example of the APP-client and APP-server).	Medium
Specification Weakness	Integrity (AP4-I)	For communication between UE and MEC as well as between MEC parts running on separate servers, the ETSI MEC specification recommends end to end encryption and integrity protection. Communication inside the virtualization layer may be protected, depending on the security needs. For authentication, certificates are recommended but not required, the decision is up to the network operator. The mechanisms are strong, so the probability that the integrity of transmitted data will be breached due to specification weaknesses is estimated to be LOW .	Low
Failure of Power Supply	Integrity (AP5-I)	An attack on the power supply or an unexpected failure can corrupt data during transmission from the MEC system to the 5G network or other data networks. Communication systems are typically designed in such a way that data transmission is secured by measures such as checksums. These are checked at the end of the transmission and the transmission is only considered correct if the check is successful. Therefore, the probability that an attack on the power supply or an unexpected failure of the power supply will falsify data to be transmitted by the MEC is estimated to be LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	In the event of a hardware failure, the affected components usually exhibit unexpected behavior. If the failure occurs during a storage process or during the transfer of data, information can be corrupted or lost. Typically, these errors are detected or corrected with technical measures such as the use of checksums or error correction. When checksums, error correction and redundant systems are used, the risk of data loss or data corruption during information transfer due to a hardware error is LOW.	Low

5.3.4 User's Identity

User data which could leak the user's identity in the context of IMSI is usually not located in the MEC, however, data for MEC specific services might be used to forge further attacks on the user.

According to the MEC standard, anonymized user identities are generally used within edge computing. In many cases, this is sufficient to distinguish the different users for the actual services. The impact of a successful attack on this identity should be limited to the MEC environment. So, this risk scenario is relevant to MEC.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The risk of leaking user's identity due to misconfiguration within the user's perspective is MEDIUM. User data which could leak the user's identity in the context of IMSI is usually not located in the MEC, however, data for MEC specific services might be. A vendor offering a MEC service might create their own database for user access and billing. Therefore, misconfiguration in a 5G MEC system, such as insecure protocol configurations, service identity spoofing (e.g., DNS), insufficient network separation (CUPS), and privilege abuse, significantly raises the risk of user identity leaks w.r.t. user's MEC identity [ESEC, BSI_Threat, ENISA]. For users accessing MEC services, these vulnerabilities can lead to their identities being misused by attackers for illegal purchases or other purposes. Insecure protocols and spoofing can intercept or redirect sensitive user information, while inadequate network separation and privilege abuse enable unauthorized access to user data [ESEC, C5, SSDP]. The probability of this risk is evaluated as MEDIUM due to the potential exploitation of these misconfigurations. Consequently, users' personal and financial information is at considerable risk, necessitating robust security measures in MEC systems to protect against such threats.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	The risk of leaking user's identity due to insufficient access control within the user's perspective is MEDIUM. User data which could leak the user's identity in the context of IMSI is usually not located in the MEC, except for specific MEC services that queried this information using possible nonstandard access via 5G Core API. This data could be easily leaked by unauthorized access and used to forge attacks on the user's identity. In the 5G MEC system, insufficient access control can lead to several attack paths that increase the probability of a user's identity being leaked. These include service identity spoofing, where attackers impersonate legitimate services to steal user data; credential abuse, where attackers exploit stolen or weak credentials to access user accounts; and malware, which can be used to capture and exfiltrate user identity information [ESEC, BSI_Threat, ENISA]. For a user accessing MEC services, these vulnerabilities mean that an attacker could misuse their identity, e.g. for illegal purchases or other purposes. In general, MEC services create a similar risk to a user as normal webservices, however, they might not be as exposed as normal webservices because of their local accessibility. The probability of such risks is assed as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	The probability that the integrity of the user can be exploited due to implementation errors or weaknesses of the MEC is considered LOW. According to the specification ETSI MEC 014 [MEC 014] and the referenced specifications, an external identifier is generated for the UE ID, which is used by the UE to identify itself to application functions (AF) as well as to the MEC and MEC applications. The mapping between the external identifier and the SUPI or IMSI takes place in the 5G core and is not visible to the MEC. As a result, attacks on the user's identity are not possible via the MEC alone. Therefore, the risk is assed as LOW. An exception would be if the MEC system itself collects user-specific login data, that is collected outside of the 5G and MEC context (login data, bank data, etc.). But this is not part of the investigation. No CVEs or exploits [CVE_Details] are known for this attack path for the MEC implementations listed by ETSI [ETSI_MEC_LIST].	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Confidentiality (AP4-C)	The protection of the 5G Identity is done by the 5G-Core. Users 5G identity cannot be accessed by MEC. MEC only has access to the AMF via the PCF and NEF. All possible interfaces to access user's data shall be restricted by NEF/PCF policies. The administration Interface shall be unreachable for MEC and MEC APPs. Charging is calculated per connection. SMF does the mapping of the connection to the UE and to the 5G User identity. SMF shall be closed for the MEC by NEF/PCF policies. The identification towards the UE is done by an anonymous external identifier. In the User Plain UPF is an IP-based Network without any links to 5G identity. So, in total with proper configuration there are no possibilities for the MEC to get knowledge of the users 5G identity. The risk of getting knowledge of the user's 5G identity within the MEC system due to specification weaknesses is LOW. Note: Charging for data volume (billing) via the 5G-Network is done as defined by 5G. How a PLMN would charge the service fees of a MEC-APP is not yet defined. Offering the charging service to an APP-Provider versus the risk of forwarding a fraudulent charge must be considered by the PLMNO. But such a charging process is considered to be outside of MEC and 5G. Usually the user needs to have a contract with an APP provider, who sells services and authenticate in a typical way, like login/password or mutual authentication.	Low
Failure of Power Supply	Confidentiality (AP5-C)	As part of the analysis, no relevant attack path was identified that could disclose the identity of one or more users in the event of a power supply failure. In the event of a total failure, a component does not grant access to its data. As it cannot be ruled out in principle that attack paths exist or will exist, the risk of a data leak is LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	As part of the risk analysis, no relevant attack path could be identified that would lead to a breach of the confidentiality of user identities if a component of multi-access edge computing were to fail. If parts of components such as the server processor or a Trusted Protection Module fail, it cannot be ruled out that the hardware error that occurs will soften security barriers introduced by design or lead to security-relevant malfunctions. Special design measures should be taken to avoid this type of error. Furthermore, it can be assumed that they occur extremely rarely and unpredictably. It is therefore difficult for potential attackers to foresee and exploit them. Furthermore, the MEC does not have access to the user database of the 5G core network but uses anonymized identities. However, providers of MEC services can set up their own user administration, which must be protected accordingly. Due to the infrequent occurrence and difficult predictability of hardware errors, the probability of the risk of a breach of confidentiality of user identities is LOW.	Low

5.3.5 Eavesdropping of a Communication

In this context, eavesdropping refers to connections between the MEC and the 5G core network, the 5G user plane and other networks connected to the MEC system. In the event of a successful attack, data exchanged via these communication interfaces can be intercepted. So, this risk scenario is relevant to MEC.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The probability of eavesdropping of a communication due to misconfigurations within the user perspective is MEDIUM. In the 5G MEC system, no data related to the core services, like phone calls or SMS, is passed through MEC services. The only possible data that can be eavesdropped is the data of, to or from the MEC services themselves. This can be metadata, such as connection information, but also user related data, depending on the actual MEC APP that is accessed by a user. Misconfigurations such as insecure protocol configurations (e.g. TLS), service identity spoofing (e.g. DNS), insufficient network separation (CUPS), missing or misconfigured certificate validation (e.g. OAuth2.0) and privilege abuse significantly increase the risk of eavesdropping on communications. [MECSEC, ESEC]. These vulnerabilities can be exploited by attackers to intercept and manipulate data traffic between the UE and MEC services. For users accessing MEC services, this means their private communications, sensitive information, and data transactions could be exposed to unauthorized parties. Therefore, the probability of the risk of a breach of confidentiality is MEDIUM.	Medium
Insufficient Access Controls	Confidentiality (AP2-C)	The probability of eavesdropping of a communication due to insufficient access control within the user perspective is MEDIUM. In the 5G MEC system, insufficient access control can lead to several attack paths that increase the risk of eavesdropping on communications. Insecure protocol configurations and insufficient network separation (CUPS) can expose user data to interception by malicious actors. Service identity spoofing, such as DNS spoofing, allows attackers to impersonate legitimate services and gain unauthorized access to sensitive information. Additionally, privilege abuse within the system can lead to unauthorized data access and eavesdropping. As there are many possibilities for an attacker to exploit insufficient access control to gain access to a user's communication data within a MEC system, the probability is evaluated as MEDIUM. For a user or UE accessing MEC services, these vulnerabilities significantly increase the probability of their communications being intercepted [CYSEC, SSDP].	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	The likelihood that confidentiality can be exploited due to implementation errors or weaknesses of the MEC is estimated to be MEDIUM. Currently there are no CVEs or exploits [CVE_Details] known, related to the MEC implementations listed by ETSI MEC Implementations [ETSI_MEC_LIST]. A classification at the MEDIUM level has still been chosen, because the assurance of the confidentiality of the communication and the measures for data access is achieved through identical methods as outlined in section 5.2.3 (TLS, OAuth 2.0, etc.). Consequently, the known risks associated with implementation weaknesses also apply here.	Medium
Specification Weakness	Confidentiality (AP4-C)	Currently, no specification weaknesses regarding eavesdropping in ETSI MEC have been published. However, [MEC-SECURITY] discusses the interaction of 5G Lawful Interception (LI) and Retained Data (RD) with MEC in detail. LI and RD are specified by 3GPP and are of course not available for the MEC. Eavesdropping would mean to duplicate IP packets and redirect the duplicates to a new IP address, similarly to LI and RD. The only way this could be done is reprogramming a switch between the UPF, MEC and the correct endpoint e.g. by setting it to debug mode, sending copies of the frames to the MEC. This may be seen as a specification weakness but also as an implementation or configuration weakness. A properly configured switch only forwards broadcast and correctly addressed frames to the MEC. Inside the MEC, the virtualization switches will take care of frame separation. Therefore, the risk of eavesdropping of communication data due to specification weaknesses is LOW.	Low
Failure of Power Supply	Confidentiality (AP5-C)	During this analysis, no relevant attack path was identified that could expose the communication data between the MEC and the 5G core network or on the data path in the event of a power supply failure. In the event of a power failure, the affected edge components fail and no longer take part in any communication. As it cannot be ruled out in principle that attack paths exist or will exist, the risk of communication being intercepted in the event of a power failure is LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	During this analysis, no relevant attack path could be identified that would lead to a breach of confidentiality in the event of a complete hardware failure. If parts of components such as the server processor or a Trusted Protection Module fail, it cannot be ruled out that the hardware error will soften security barriers introduced by design or lead to security-relevant malfunctions. Special design measures should be taken to avoid this type of error. It can therefore be assumed that they occur extremely rarely and unpredictably. It is difficult for potential attackers to foresee and exploit them. For this reason, the risk of a breach of confidentiality is LOW.	Low

5.3.6 Slowing Down/Restricting the Connection

For MEC solutions in a public 5G network, a deterioration in the quality of service means that, for example, the required latencies of connections between user equipment and a MEC service for an "autonomous driving" use case cannot be met. In the worst case scenario, measures to prevent a car accident are initiated too late.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The probability of slowing down / restricting the connection due to misconfiguration within the user perspective is HIGH. In the 5G MEC system, misconfigurations such as insufficient DoS resilience, poor resource planning, and inadequate network separation can significantly slow down or restrict user connections. If the system lacks proper defense against DoS attacks [DOSATT], it becomes vulnerable to being overwhelmed, leading to degraded service performance. Poor resource planning fails to accommodate load peaks, resulting in service bottlenecks during high demand [MECCHAL]. Inadequate network separation in the Control and User Plane Separation (CUPS) can cause interference and traffic congestion. Appropriate resource planning is considered a crucial task for MEC hosts, as they are most likely subject to resource constraints (limited rack space, power, network connection, etc. at the edge location). Peak load situations and the resulting computing and networking requirements are in some cases hardly predictable (local events, road traffic, etc.). For users accessing MEC services during peak times, these issues can lead to frustratingly slow or unavailable connections. The probability of such risks occurring is HIGH due to the complexity of system dimensioning.	High
Insufficient Access Controls	Availability (AP2-A)	The probability of slowing down / restricting the connection due to insufficient access control within the user perspective is LOW. In a 5G MEC system, insufficient access control can lead to attack paths such as insufficient resource capacity limits and credential/identity abuse [SUPCHAIN]. These vulnerabilities can slow down or restrict the connection for users accessing MEC services. For instance, inadequate resource limits might cause resource exhaustion, reducing the availability and performance of services. Additionally, credential abuse can allow unauthorized users to monopolize resources, further degrading service quality [CYSEC]. From a user's perspective, these issues translate to poor service reliability and increased latency, which is especially impactful if real time capabilities are needed. However, a malicious actor or service exploiting insufficient access control might most likely lead to a complete failure of the system, rather than a slow or restricted connection. Therefore, the probability of a slow connection occurring due to insufficient access control is evaluated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Despite the increasing threat of DoS attacks and the existence of numerous implementation weaknesses in protocols that are susceptible to such attacks, the risk is classified as LOW. The reason for this is that a MEC will be accessible only to a limited number of participants, as it operates locally, in contrast to servers on the internet that are accessible to many users and possible attackers. Since a MEC system is typically located close to the UE, it is assumed that the attack cannot be carried out by many connected bots, as would be the case with direct internet accessibility of a service. Vulnerabilities in protocols and specifically parsers [JSON, XML] generally present an attack surface for DoS, for example OpenSSL buffer overflow [CVE-2024-5535], IPSec heap overflow [CVE-2024-21894], insufficient input/output validation [OWASP_M4] and many more. It should be considered separately if DoS attacks, or insufficient resource management prevent the required latency times from being met; this should be checked through appropriate penetration tests. Weaknesses in resource management or weaknesses in the implementation of services and applications are considered more likely to impair availability. The system should be tested for robustness against poorly programmed applications.	Low
Specification Weakness	Availability (AP4-A)	Current specification concentrates on the ETSI MEC-framework, the virtualization, the interfaces to 5G and the separation of MEC APPs. The 5G cores local AMF and the local UPF are closest edge of the edge computing towards the UEs. As MEC and 5G can be configured to reserve bandwidth for different purposes (network slices), separate instances of MEC can be used with different characteristics (URLLC, eMBB) besides each other with granted behavior. So, the risk of slowing down/restricting the connection due to specification weaknesses is LOW. Note: In industrial scenarios where private 5G networks are used, company specific solutions will dominate. Their edge may be deployed at the distributed unit (DU) of the RAN, offering lowest latency with reserved processing power and bandwidth. This may slow down the non-MEC system significantly. On the other hand, an industrial system will use the MEC technology to reduce the bandwidth between the DU to the CU. Monitoring will show bottlenecks, which can be taken care of by adding hardware resources or configuration. An example would be a PLC APP running on the DU-server and interacting with its inputs and outputs directly via the RAN.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The physically less protected areas in the edge make it easier to attack the power supply directly in order to take edge servers and edge components out of operation. In small systems without redundancy, this leads to a total failure of services; in distributed systems, the failure of individual computing resources leads to a slowdown of services. Redundant structures in the power supply can prevent failure. If this mechanism does not take effect, a failure is usually detected quickly, and countermeasures can be taken. In addition, the probability can be further reduced by securing edge servers against unauthorized physical access in accordance with the current state of the art [SEC_EMBED]. Therefore, the probability that manipulation of the power supply or an unexpected failure of the power supply will slow down or restrict a connection to and from the MEC is LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	One possible failure mechanism is that edge components are often operated in non-air-conditioned and physically less protected environments. In these locations, there is a risk that environmental influences [ENV_INFLU] or the effects of physical force will disrupt the operation of the hardware or destroy the component. The use of redundancy in sensitive areas can reduce the risk that the failure of individual computing resources leads to a slowdown or, in the worst case, to a connection failure. The probability can be further reduced by additionally securing edge servers against unauthorized physical access according to the current state of the art [SEC_EMBED]. Therefore, the probability that a hardware failure or attack on the hardware slows down or restricts a connection to and from the MEC is LOW.	Low

5.3.7 Connection Failure

Similarly to the risk scenario "Urgent Telephone Calls Not Possible", multi-access edge computing cannot influence telephone calls.

Multi-access edge computing is therefore not considered part of the "Connection Failure" risk scenario.

5.3.8 Tapping or Altering of the Geographical Position

This scenario simultaneously addresses confidentiality, if an attacker can access information on the current geographical position, and integrity of the position information if he manages to change it. A separate location API is specified for MEC, which provides services with location information of the individual end devices. This means that corresponding information on the current position can be stored in the end device, in the core network and in the MEC. So, this risk scenario is relevant to MEC.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	The risk of tapping or altering the geographical position due to misconfiguration within the user perspective is LOW. Like “Telecommunication Secrecy” - Geographical data is a specific type of user data which could be tapped or altered in a system. Therefore, a similar probability assessment as in “Telecommunication Secrecy” is applied. In a 5G MEC system, misconfiguration can lead to several attack paths. Privilege abuse, resulting from not adhering to the principle of least privilege, allows malicious users access, enabling further attacks on MEC components and compromising confidentiality of user location data. Service identity spoofing, due to misconfigured authentication and inadequate API verification, can redirect users to malicious services, leaking user location data. Insecure protocol configurations, often from using default values and outdated protocols, lead to insecure data transfers and potential identity spoofing, with misconfigured logging allowing attackers to remain undetected [ESEC, SSDP, MSHAF, OWASPTLS]. Vulnerabilities enable attackers to tap or alter the geographical position data by masquerading as legitimate network functions or services, abusing user credentials, or exploiting misconfigured protocols. For a user or UE accessing MEC services, this means that their location data could be intercepted, modified, or misused without their knowledge, leading to privacy breaches and potential security threats. However, the MEC system itself does not track and maintain the geographical location of users. A MEC service can request UE locations from the client service on the UE, or from the standardized MEC Location API, which retrieves the information from the 5G core. This would enable critical services to compare location information provided by the UE application and the 5G core. The relevant MEC Location API for UE location retrieval is designed to be a read-only API (i.e., only HTTP-GET method is supported, as specified for the MEC location API [MECLOCAPI]), so the risk of altering data is considered as LOW and can additionally be mitigated by information fusion from different sources.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Integrity (AP2-I)	The risk of tapping or altering the geographical position due to insufficient access control within the user perspective is LOW. Like “Telecommunication Secrecy” - Insufficient access control can lead to significant risks such as malware infections and service identity spoofing. Malware can infiltrate the MEC platform, compromising confidentiality, integrity, and availability by exfiltrating location data, and enabling further attacks. Similarly, lack of proper authentication can allow evil services to impersonate legitimate ones, intercepting and corrupting location data transmissions. For a user, this poses a high risk to its location data, as sensitive communications could be intercepted and manipulated [CYSEC, SSDP]. In the scope of MEC systems, the attack vectors for tapping or altering a geographical location of a user are relatively difficult to fulfil. The probability of this risk is therefore evaluated as LOW.	Low
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	The probability that the confidentiality of geographical position data will be compromised due to implementation weaknesses is generally considered LOW in terms of the risk scenario. The statements on "eavesdropping of communications", which generally classify a breach of data secrecy as MEDIUM, also apply here, but for the gap to be exploited in the context of the risk scenarios, it must also be possible to infer the UE and the user. In the "User's Identity" risk scenario, inferences about the user were rated as LOW, so the overall risk assessment here is considered also LOW. However, additional user data that an APP provider collects for the use of its APP (navigation or routing) and does not adequately protect is not part of the investigation. No CVEs or exploits [CVE_Details] are currently listed for the MEC implementations listed by ETSI [ETSI_MEC_LIST].	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Integrity (AP4-I)	Tapping the location of a UE via the AMF location service in 5G is widely used to track citizens in foreign countries [FINDING_YOU]. The geographical position due to the current 5G specification is read from the Location management function (LMF) via the AMF. The interface is generally available for any AF. If it was made available by NEF and PCF for the MEC, transport and storage of the data should be protected for confidentiality and integrity. The access rights inside the MEC are handled per MEC APP and controlled by OAuth2. Digital signatures for authentication are possible. The specification recommends protection, but it is up to the network operator, to deal with it. As the position data accuracy is 0,1 to 15m depending on the wavelength and antenna types, tracking of these data over a day may reveal the identity of the user. Therefore, the use of this interface should either be disabled by policies, or the access rights should be restricted to trusted APPs inside the MEC. However, this is not a specification weakness, but rather a configuration decision. The user has to rely on the configuration, but the MEC specification defines measures to build a secure system. The risk of tapping the geographical position due to specification weakness within the user perspective is therefore seen as LOW. The risk of altering of the geographical position due to specification weakness within the user perspective is also seen as LOW. Note: Forwarding the GPS-Data to MEC APPs is under the control of the user. This mechanism is not in the scope of this document.	Low
Failure of Power Supply	Integrity (AP5-I)	In the event of a power supply failure, there is a risk that data and thus geographical location information will be corrupted. The probability of damage is LOW, as in local campus networks, for example, consistency checks can be used to monitor for jumps in the location information. This is more difficult in public networks, as UEs may be switched off during a flight, for example. If the position data is stored in a database, the database can be secured with consistency checks. Therefore, the probability that an attack on the power supply or an unexpected failure of the power supply will falsify the stored or transmitted geographical location information of a subscriber within the MEC is LOW.	Low
	Confidentiality (AP5-C)	As part of the analysis, no relevant attack path was identified that could expose the communication data between the MEC and the 5G core network or on the data path in the event of a power supply failure. In the event of a power failure, the affected edge component fails and no longer takes part in any communication. As it cannot be ruled out in principle that attack paths exist or will exist, the risk of unauthorized reading or interception of position information in the event of a power failure is LOW.	

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	In the event of a hardware failure, it should be noted that the affected components usually fail spontaneously or exhibit unexpected behavior. This can happen during a storage process or during the transfer of data. In this case, regarding integrity, position data can be falsified or lost. Measures such as checksums can be used to detect this or correct errors. However, the bigger problem is the failure of the respective component. Security components such as TPMs should be designed in such a way that data is not falsified or disclosed in the event of a hardware failure. Incorrect behavior due to design errors cannot be ruled out. For example, attackers gained access to AMD TPM modules with side-channel attacks [AMD_TPM]. Design errors could be the reason for this. If checksums, error correction and redundant systems are used, the risk of data loss or data corruption when retrieving and saving position data due to a hardware error is LOW.	Low
	Confidentiality (AP6-C)	During this analysis, no relevant attack path could be identified that would lead to a breach of the confidentiality of position data if a hardware component were to fail. If parts of components such as the server processor or a Trusted Protection Module fail, it cannot be ruled out that the hardware error that occurs will soften security barriers introduced by design or lead to security-relevant malfunctions. Special design measures should be taken to avoid this type of error. It can therefore be assumed that they occur extremely rarely and unpredictably. It is difficult for potential attackers to foresee and exploit them. For this reason, the probability of the risk of a breach of confidentiality of position data is LOW.	

5.4 Private 5G Network Operator Perspective

Note: The following assumptions apply for these risk scenarios for the private 5G network perspective.

- This scenario is only focussed on on-premise networks.
- The network's backbone is part of the organization's general IT network.
- The network is separated from the public network by a firewall, which is part of the 5G network.
- The network does not have an IP Multimedia Subsystem (IMS), but uses Multi-Access Edge Computing.
- This perspective considers networks that are limited to dedicated locations.
- Organizations might operate multiple locations with separate private 5G networks.
- The members of the organization operating the network are also the user of the network. There is no separate group of users or a separate operator.

5.4.1 Network Failure

The triggering of a network error within a private 5G network can lead to a restriction or failure of MEC services. In the area of a private network used in an industrial context, this can affect other communication interfaces in addition to 5G, e.g. the operational technology in the production area. So, this risk scenario is relevant to MEC.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The probability of network failure due to misconfiguration of the MEC system from the perspective of a private network operator is considered MEDIUM. Like “Failure of Mobile Service” in the public MNO perspective - Misconfiguration issues in a private network operators MEC environment, such as insufficient resource planning, poor network separation, or errors in network configuration (DNS, routing tables) can impact the risk of a network failure. Insufficient resource planning or a wrong network configuration can lead to outages of services deployed on a MEC system and degraded performance, affecting especially applications that depend on low latency and high availability. Poor network separation can create security vulnerabilities, allowing unauthorized access and interference, potentially disrupting service integrity and availability. The root cause of these issues is inadequate configuration or planning. The probability of this risk is considered MEDIUM because misconfiguration is a very likely root cause for mobile services failure [ENISA, ESEC], in this case specifically for MEC services. In addition, network administrators of small- and medium-sized companies may not have the experience, full-depth knowledge, or applied workflows to mitigate the risk of misconfigurations and subsequent network failures.	Medium
Insufficient Access Controls	Availability (AP2-A)	The probability of a network failure due to insufficient access control from the perspective of a private network operator is LOW. Insufficient access control, such as lack of resource capacity limits and improper credential management, can lead to malicious services overloading a MEC platform, causing MEC service failure. Additionally, weak authentication practices can result in credential abuse, allowing attackers to set malicious configurations, rendering the MEC mobile service unavailable [ENISA, SSDP]. For a private network operator, these issues can increase the risk of network failure, disrupting user experience and service availability. However, due to the restricted access to the 5G network, the 5G network operation in/behind a protected corporate network (i.e., behind a firewall), and well-defined scoped of the applications and the resources that are deployed on a MEC system, the probability of this risk is LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	In a private 5G network, the probability of a network failure due to implementation flaws of MEC components is considered as LOW. Although MEC relies on the services of the 5G network and may contain potential vulnerabilities, the likelihood that the MEC system itself will cause a failure of the private network is LOW, even if implementation flaws can never be ruled out. The classification LOW is based on the fact that the framework conditions and actual requirements in operation are better known than in a public network and MEC APPs as well as MEC Services can be better tested against the requirements they will face in productive operation as in a public network. Compared to a public network, it is rather unlikely that unexpected events will occur. For example, it is unlikely that a large number of user end devices (UEs) will suddenly dial in to a cell or that there will be unexpectedly high bandwidth requirements. The system should therefore better match the requirements than a public network, as these are better known and the network can be designed accordingly during the design phase to fulfill the requirements. Furthermore, the MEC platform is operated isolated in a private network and is therefore protected against external access.	Low
Specification Weakness	Availability (AP4-A)	Erroneous or attacking APPs may overload their own container or other APPs. A failure of the MEC in total is prevented by virtualization and orchestration as specified. Logging and alerts allow a fast identification and elimination of the bad APP. So, the specification defines mechanisms to prevent these threats. With a proper configuration of policies and a stable 5G-core, the probability of a failure of mobile services is LOW.	Low
Failure of Power Supply	Availability (AP5-A)	The physically less protected area in the edge makes it easier to attack the power supply directly to take edge servers and edge components out of operation. The risk can be reduced, if there are redundant structures in the power supply that prevent a failure. If this mechanism does not take effect, a failure inside a private network is usually detected quickly, and countermeasures can be taken. In addition, the probability can be further reduced by securing edge servers against unauthorized physical access. The risk of a power failure can be reduced by using an uninterruptible power supply (UPS) or an emergency power generator. This is much easier to implement in private networks due to the geographical proximity of the edge components to each other. If the measures described are used, there is a LOW probability that manipulation of the power supply or an unexpected failure of the power supply to one or more edge components in a private network will lead to network failures.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Edge components are often operated in non-air-conditioned and physically less protected environments. There is a risk that environmental influences [ENV_INFLU] or physical force may disrupt the operation of the hardware or destroy the component. The use of redundancy in sensitive areas can reduce the risk that the failure of individual computing resources will lead to a slowdown or, in the worst case, to a connection failure. The probability can be further reduced by additionally securing edge servers against unauthorized physical access in accordance with the current state of the art [SEC_EMBED]. A special feature of private networks, such as an industrial campus network, is that industrial PCs, industrial controllers and embedded systems such as a Raspberry PI work as edge components alongside servers. These components generally have no redundancy in operation but are replaced with replacement hardware held in stock in the event of a hardware failure to minimize potential downtime. Using the measures described, the probability that a hardware failure or an attack on the hardware will lead to a network error or failure in a private network is LOW.	Low

5.4.2 Degraded Services / Connections

Failures of important computing and communication resources in the MEC area can lead to a reduction in the quality of service regarding services and connections. If resources are missing, there is a very high probability that the effects of an attack can lead to a degradation of services or connections. This can lead to high costs and reputation-damaging delivery delays, especially in the production sector, where, for example, exact recipes or latencies in communication connections must be adhered to. So, this risk scenario is relevant to MEC.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	The risk of degraded services / connections due to misconfigurations from the perspective of a private network operator is MEDIUM. Like “slowing down / restricting the connection” in the User Perspective: In the 5G MEC system, misconfigurations such as poor resource planning, insufficient DoS resilience, and inadequate network separation can significantly slow down or restrict user connections. Poor resource planning fails to accommodate load peaks, resulting in service bottlenecks during high demand [MECCHAL]. If the system lacks proper defense against DoS attacks [DOSATT], it becomes vulnerable to being overwhelmed, leading to degraded service performance. Inadequate network separation in the Control and User Plane Separation (CUPS) can cause interference and traffic congestion. Appropriate resource planning is considered a crucial task for MEC hosts also in private network installation, as they are most likely subject to resource constraints (limited rack space, power, network connection, etc. at the edge location). Peak load situations and the resulting computing and networking requirements remain challenging, but compared to public networks, private networks are more plannable environments as the requirements of applications are usually well-known beforehand such that resources can be implemented accordingly. For users accessing MEC services during peak times, these issues can still lead to frustratingly slow or unavailable connections. The probability of such risks occurring is MEDIUM, due to the complexity of system resource dimensioning. Considering the usually well-defined requirements of an on-premises network, the deployed applications, and the number of users, the probability is not evaluated as high.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Availability (AP2-A)	The risk of degraded services / connections due to insufficient access control from the perspective of a private network operator is LOW. Like “slowing down / restricting the connection” in User Perspective: In a 5G MEC system, insufficient access control can lead to attack paths such as insufficient resource capacity limits and credential/identity abuse [SUPCHAIN]. These vulnerabilities can slow down or restrict the connection for users accessing MEC services. Additionally, credential abuse can allow unauthorized users to monopolize resources, further degrading service quality [CYSEC]. From a user's perspective, these issues translate to poor service reliability and increased latency, which is especially impactful, if real time capabilities are needed. However, due to the usually well-defined scope of the applications running on a MEC system and the restricted access to private 5G networks, i.e. located behind a firewall, the probability of this risk is evaluated as LOW.	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	In a private 5G network, the probability of a network failure due to implementation flaws of the MEC is considered as LOW. Currently, there is no reliable information on implementation flaws in MEC systems and their services and applications available. Nevertheless, potential software implementation errors that could lead to a degradation of services and connections must also be expected in this context, like DoS attacks for example or implementation flaws that lead to buffer overflows or overload situations [CVE_Details]. In contrast to public networks, however, the operator has better options for selecting vendors of MEC services and APPs. In addition, continuous monitoring can reduce the likelihood of significant outages. Therefore, the probability of this risk is rated as LOW.	Low
Specification Weakness	Availability (AP4-A)	Specification includes mechanisms to grant different levels of processing power to a service. This is supervised by the virtualization and orchestration. Overload situations are still possible, as hardware resources are limited in a typical MEC scenario regarding private networks. With proper policies, an overload of the MEC may temporarily reduce the bandwidth to other connections, but in a controlled way. In general, the specification allows to distribute the resources in a predictable way to the various services and APPs. As there are no publicly known specification weaknesses regarding this scenario, the probability is for Degraded Services/Connections due to specification weakness is rated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	The physically less protected area in the edge makes it easier to attack the power supply and take edge servers and edge components out of operation and, depending on the impact, leads to a deterioration in the quality of service of MEC services or connections up to failure. The risk can be reduced, if there are redundant structures in the power supply that prevent a failure. If this mechanism does not take effect, a failure inside a private network is usually detected quickly, and countermeasures can be taken. In addition, the probability can be further reduced by securing edge servers against unauthorized physical access. The risk of a power failure can be reduced by using an uninterruptible power supply (UPS) or an emergency power generator. This is much easier to implement in private networks due to the geographical proximity of the edge components to each other. If the measures described are used, there is a LOW risk that manipulation of the power supply or an unexpected failure of the power supply to one or more edge components in a private network will lead to a degradation of services or connections.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Edge components are often operated in non-air-conditioned and physically less protected environments. There is a risk that environmental influences [ENV_INFLU] or physical force may disrupt the operation of the hardware or destroy the component. The use of redundancy in sensitive areas can reduce the risk that the failure of individual computing resources will lead to a slowdown or, in the worst case, to a connection failure. The probability can be further reduced by additionally securing edge servers against unauthorized physical access in accordance with the current state of the art [SEC_EMBED]. A special feature of private networks, such as an industrial campus network, is that industrial PCs, industrial controllers and embedded systems such as a Raspberry PI work as edge components alongside servers. These components generally have no redundancy in operation but are replaced with replacement hardware held in stock in the event of a hardware failure to minimize potential downtime. Using the measures described the probability that a hardware failure or an attack on the hardware will lead to a deterioration in the quality of service for MEC services and connections is LOW.	Low

5.4.3 Information Disclosure

In private 5G networks and in the area of MEC, there is a great need for protection against the disclosure of confidential information. In addition to the forementioned case of the publication of medical and personal data handled within the MEC, the disclosure of secret company data such as production processes and secret formulas can lead to existential problems for companies. So, this risk scenario is relevant to MEC.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	The probability of information disclosure due to misconfigurations from the perspective of a private network operator is MEDIUM. Like “Telecommunication Secrecy” in the public MNO perspective: Privilege abuse, resulting from not adhering to the principle of least privilege, allows malicious users access, enabling further attacks on MEC components and compromising confidentiality of user data. Service identity spoofing, due to misconfigured authentication and inadequate API verification, can redirect users to malicious services, leading to information disclosure by leaking data transmitted by the user. Insecure protocol configurations, often caused by using default values and outdated protocols, lead to insecure data transfers and potential identity spoofing, with misconfigured logging allowing attackers to remain undetected [ESEC, SSDP, MSHAF, OWASPTLS]. For a private network operator, these attack paths collectively threaten information disclosure by exposing sensitive data and disrupting service integrity and availability. Private networks are operated most likely in/behind a corporate network (i.e., behind a firewall), so administrators and developers may trust in this barrier and do not protect or configure security mechanisms like they would for public internet systems (a robust and secure infrastructure needs a multitude of security measures to be effective). Consequently, the probability of this risk is rated as MEDIUM.	Medium
Insufficient Access Controls	Confidentiality (AP2-C)	The probability of information disclosure due to insufficient access control from the perspective of a private network operator is LOW. Like “Telecommunication Secrecy” in the public MNO perspective: Insufficient access control can lead to significant risks such as malware infections and service identity spoofing. Malware can infiltrate the MEC platform, compromising confidentiality, integrity, and availability by exfiltrating sensitive data, disrupting services, and enabling further attacks. Similarly, lack of proper authentication can allow evil services to impersonate legitimate ones, intercepting and corrupting data transmissions. For a private network operator, this poses a high risk of information disclosure, as sensitive communications could be intercepted and manipulated [CYSEC, SSDP]. Due to the restricted access to the private 5G network, i.e. located behind a firewall, the known operated applications, and the trusted user group, the probability of this risk is LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	In a private 5G network, the probability of an information disclosure due to implementation flaws of the MEC is considered as LOW. Currently, there are no reliable information regarding implementation flaws in MEC systems available with regards to information disclosure. Nevertheless, the weaknesses and flaws in the implementation of protocols for data confidentiality and access protection, as listed in Section 5.3.2 "Telecommunication Secrecy" are also applicable here. The classification of LOW is chosen because the MEC in a private 5G network is accessible only to a group of trusted users and is protected against external attacks through additional measures, including isolation. To ensure additional protection against internal attackers, sensitive data should be stored in encrypted form on storage media.	Low
Specification Weakness	Confidentiality (AP4-C)	In a private network, the set of MEC APPs is well selected because of its industrial application. The APPs will rather be company specific or application specific, specially programmed for the factory. Untested third-party APPs will be banned. Still all the mechanisms are available to protect stored information against disclosure like authentication via certificates, OAuth2, end-to-end encryption and integrity protection. The ETSI MEC specification refers also to virtual machine encapsulation for higher security or special hardware for key storage (TPM, Smartcard, USIM). So, the combination of lower security requirements and proper security measures as specified by ETSI MEC leads to a LOW probability that such information is disclosed. Most likely, non-public (company specific) specification is used, so weaknesses cannot be pointed out.	Low
Failure of Power Supply	Confidentiality (AP5-C)	As part of the analysis, no relevant attack path could be identified that would lead to the disclosure of confidential information in the event of a power supply failure in a private network. As it cannot be completely ruled out that attack paths exist or will exist, the probability of a breach of confidentiality in a private network is LOW.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	As part of the risk analysis, no relevant attack path could be identified in a private network that would lead to the disclosure of confidential information in the event of a complete hardware failure. If parts of components such as the server processor or a Trusted Protection Module fail, it cannot be ruled out that the hardware failure that occurs will soften security barriers introduced by design or lead to security-relevant malfunctions. Special design measures should be taken to avoid this type of error. It can therefore be assumed that they occur extremely rarely and unpredictably. It is difficult for potential attackers to foresee and exploit them. For this reason, the probability of a breach of confidentiality of relevant information in a private network is LOW.	Low

5.4.4 User Identities (Authenticity/Integrity)

An attack on the integrity of user identities can lead to various problems in the area of MEC. According to the MEC standard, anonymized user identities are generally used within edge computing. In many cases, this is sufficient to distinguish between the different users for the actual services. The impact of a successful attack on this identity should be limited to the MEC environment. However, it can be assumed that the safety behavior of robots and transport vehicles works autonomously and cannot be attacked. Attacks can, however, create situations that massively disrupt production processes. For example, industrial Automated Guided Vehicles (AGV) can be steered into an area where they interfere with each other, and transportation comes to a standstill. So, this risk scenario is relevant to MEC.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	The probability of insufficient integrity and authenticity of a user's identity due to misconfiguration from the perspective of a private network operator is LOW. User data which could leak the user's identity in the context of IMSI is usually not located in the MEC, however, data for MEC specific services might be. A vendor offering a MEC service might create its own database for user access and billing. Therefore, misconfiguration in a 5G MEC system, such as insecure protocol configurations, service identity spoofing (e.g., DNS), insufficient network separation (CUPS), and privilege abuse, significantly raises the risk of user identity leaks w.r.t. user's MEC identity [ESEC, BSI_Threat, ENISA]. For users accessing MEC services, these vulnerabilities can lead to their identities being misused by attackers for illegal purposes. Insecure protocols and spoofing can intercept or redirect sensitive user information, while inadequate network separation and privilege abuse enable unauthorized access to user data [ESEC, C5, SSDP]. However, as private networks are operated in/behind corporate networks, a misconfiguration of a private MEC service may not directly expose the vulnerability to an external attacker. MEC systems use well-known and robust protocols for authentication and authorization such as e.g. OAuth2.0 [IETFOAUTH]. Known attacks are more related to phishing attacks than to a technical problem with the method [OAUTHGOOGLE]. Therefore, the probability of this risk is evaluated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Integrity (AP2-I)	The probability of insufficient integrity and authenticity of a user's identity due to insufficient access control from the perspective of a private network operator is MEDIUM. In a 5G MEC system, insufficient access control can lead to significant risks concerning user identity authenticity and integrity. Attack paths such as network function identity spoofing and service identity spoofing, e.g. due to insufficient user authentication, allow attackers to impersonate legitimate network functions or services, potentially leading to unauthorized access to user data and services. Credential or identity abuse enables attackers to misuse or steal credentials, further compromising user identities [ESEC, ENISA, SSDP]. For a private network operator, these vulnerabilities can result in severe breaches of user trust and service disruption. In addition, in corporate environment also special user profiles exist, such as function users for machines or with shared passwords. Additionally, security measures might be implemented with respect to a trusted environment (e.g., missing user roles or access right levels), creating another risk factor for unallowed access and user identity spoofing. The reported attack vectors are critical and therefore increase the probability of users' identities being misused. However, due to the limited attack surface of on-premise networks and MEC systems, the probability is rated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	In a private 5G network, the probability of user identity theft due to implementation flaws of the MEC is considered as MEDIUM. According to the MEC standard, anonymized user identities are generally used within edge computing. The actual identity of a user or an end device therefore remains hidden. TLS can be used to secure the identity at the MEC APP level, but there are no binding specifications for this in the private network. Due to some known implementation weaknesses of TLS as well as token-based procedures and implementation errors in certificate checks (acc. section 5.2.3), the MEDIUM level is chosen.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Integrity (AP4-I)	While in a PLMN the 5G identity is directly linked to the user, this is different in private 5G networks that are used in an industrial scenario. Here the USIMs are generated by the operator of the private network and most UEs will be machines or sensors and actors. The user identity may be a unique number generated at production time. The human users' name or description may be stored separately. The description of a machine UE contains usually its type, function, position, address and machine unit. In a production plant a person has to subscribe to a contractual document concerning the safety rules, and in such an environment people need to be locatable permanently for safety reasons. The UE identity is the IP-address of the UE. As mentioned above, by using TLS and certificate chains, the APP can verify the IP-address and access rights of the user. Still, a separation of USIM based identity from the MEC can be configured by the network operator. In current discussions, 5G sensors are seen as low-cost products, with low memory resources that cannot hold a complex security API and may therefore act unprotected, only based on IP-addresses. ETSI MEC does not enforce encryption, neither does the RAN. Still, this is an ongoing discussion. Overall, the probability of insufficient integrity and authenticity of a user's identity due to specification weakness is seen LOW .	Low
Failure of Power Supply	Integrity (AP5-I)	An attack on the power supply or an unexpected failure can corrupt the anonymized user identity during transmission between different components of a MEC system in a private network. Communication systems are typically designed in such a way that data transmission is secured by measures such as checksums or error correction. The same applies to the storage of information about a user in databases or a file system within multi-access edge computing. The consistency of storage media or the violation of integrity can be ensured by measures such as the use of checksums or error correction. Therefore, the probability that an attack on the power supply or an unexpected failure of the power supply will jeopardize the integrity of user identities in a private network is LOW .	Low
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	In the event of a hardware failure, the affected components usually exhibit unexpected behavior. If the failure occurs during a storage process or during the transfer of data such as user identities, this information can be corrupted or lost. Errors can be detected or corrected with technical measures such as the use of checksums or error correction, e.g. by using redundant systems and comparing the statuses. There are no differences in the procedure between multi-access edge computing systems in public 5G networks and private 5G campus networks. When checksums, error correction and redundant systems are used, the probability of data loss of user identity or data corruption during information transmission or storage due to a hardware error in private networks is LOW .	Low

6 Risk Assessment Results

This chapter presents the final results of the detailed risk analysis.

Note: Risk scenarios with no given probability level were not rated, see beginning of Chapter 5 and the corresponding risk scenario.

6.1 Probability Assessment

The following table summarizes the results of this detailed analysis of the **MEC system**. According to the maximum principle in risk analysis only the highest probability is listed.

Perspective	Section	Risk Scenario	Probability Level
Public MNO	5.2.1	Emergency Call (TKG § 164)	
	5.2.2	Cell Broadcast (TKG § 164a)	
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	Medium
	5.2.4	Emergency Preparedness (TKG § 184)	
	5.2.5	Regional Network Failure	
	5.2.6	Multi-Regional Network Failure	
	5.2.7	Failure of Mobile Service	High
	5.2.8	Fraud	
	5.2.9	Subscriber-Data Leak	
User	5.3.1	Urgent Telephone Calls Not Possible	
	5.3.2	Authenticity of the Receiver of a Call	
	5.3.3	Integrity of Transmitted Data	Medium
	5.3.4	User's Identity	Medium
	5.3.5	Eavesdropping of a Communication	Medium
	5.3.6	Slowing Down/Restricting the Connection	High
	5.3.7	Connection Failure	
	5.3.8	Tapping or Altering of the Geographical Position	Low
Private 5G Network Operator	5.4.1	Network Failure	Medium
	5.4.2	Degraded Services/Connections	Medium
	5.4.3	Information Disclosure	Medium
	5.4.4	User Identities (Authenticity/ Integrity)	Medium

6.2 Risk Assessment

The following table summarizes the overall risk assessment of the **MEC system**. The risk scenarios and their respective impact level are described in the framework document.

Perspective	Risk Scenario	Impact	Probability	Risk Score
Public MNO	Emergency Call (TKG § 164)			
	Cell Broadcast (TKG § 164a)			
	Telecommunications Secrecy (TTDSG § 3)	Medium	Medium	Medium
	Emergency Preparedness (TKG § 184)			
	Regional Network Failure			
	Multi-Regional Network Failure			
	Failure of Mobile Service	Medium	High	High
	Fraud			
	Subscriber-Data Leak			
User	Urgent Telephone Calls Not Possible			
	Authenticity of the Receiver of a Call			
	Integrity of Transmitted Data	High	Medium	High
	User's Identity	High	Medium	High
	Eavesdropping of a Communication	Medium	Medium	Medium
	Slowing Down/Restricting the Connection	Low	High	Medium
	Connection Failure			
	Tapping or Altering of the Geographical Position	High	Low	Medium
Private 5G Network Operator	Network Failure	Medium	Medium	Medium
	Degraded Services/Connections	Medium	Medium	Medium
	Information Disclosure	Medium	Medium	Medium
	User Identities (Authenticity/ Integrity)	Medium	Medium	Medium

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